

Dynamics of a two-patch epidemic model with deterministic/stochastic migration and distributed delays

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ABSTRACT

We study the global dynamics of a two-patch epidemic model that integrates spatial migration, Erlang-distributed delays, and environmental stochasticity. The two patches are coupled by migration whose intensities are modulated by an Ornstein-Uhlenbeck (OU) process, capturing stochasticity but mean-reverting fluctuations in population movement, while the infection progression is described by a distributed delay structure represented through a linear-chain formulation. For the deterministic system, we establish threshold-type results that characterize disease extinction and persistence: the disease-free equilibrium is globally asymptotically stable when the basic reproduction number $\mathcal{R}_0 < 1$, whereas infection persists when $\mathcal{R}_0 > 1$. For the corresponding stochastic model, we derive explicit sufficient conditions for almost sure exponential extinction by constructing Lyapunov functions and exploiting the Metzler structure of the infected subsystem. Moreover, we prove that the stochastic system admits a stationary distribution when stochastic threshold $\mathcal{R}_0^s > 1$, which serves as a stochastic analogue of the endemic equilibrium and describes persistent random fluctuations of infection levels. Numerical simulations are provided to validate the theoretical findings and to quantify how distributed delays and OU-driven migration randomness shape long-term infection burden and stationary prevalence across patches. Notably, adjusting the noise intensity and the mean-reversion rate can redistribute the long-term infection burden between patches, which highlights how migration-driven randomness fundamentally reshapes spatial epidemic patterns.

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1. Introduction

Since Kermack and McKendrick first established the compartment model (Kermack & McKendrick, 1927), differential equation systems have become the core tool for analyzing the transmission mechanism of infectious diseases, predicting the trend of epidemics and formulating prevention and control measures (Allen, 2008; Bai et al., 2025; Hethcote, 2000; Song &

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Song, 2025; Zhang et al., 2026). Among various infectious disease models, the patch model has been proposed to extensively study how individual mobility and spatial heterogeneity, such as commuting, trade or transportation, affect the dynamics of the disease (Alrawas et al., 2024; Bai et al., 2026; Gou et al., 2023; Huang et al., 2025; Liu & Chen, 2025; Lou, 2019; Shi et al., 2022; Shuai & van den Driessche, 2014). Furthermore, the evolution of the epidemic does not occur instantaneously: from coming into contact with the source of infection to undergoing testing, from receiving a positive test result to becoming contagious, each step involves a time delay (Feng et al., 2026; Kiouach & Sabbar, 2021; Liu et al., 2025; Ratnayake et al., 2020; Yu et al., 2019). These seemingly insignificant time delays actually significantly alter the transmission pattern of the epidemic and the development trend of the disease. Therefore, time delay must be explicitly incorporated into the model to be reflected. In general, two primary types of delay are considered in mathematical epidemiology: discrete delays (Feng et al., 2026; Moujahid & Vadillo, 2026; Yu et al., 2019) and distributed (or continuous) delays (Kiouach & Sabbar, 2021; Liu et al., 2025; Zhang, 2025). Discrete delays refer to the fact that the behaviors of the system depend on the state at one or more specific historical moments, such as a constant incubation period from coming into contact with the source of infection to the onset of infection. It can simplify complex biological processes into clear lag relationships, which are usually easier to handle mathematically. However, biological and behavioral processes in the real world often have individual differences. For example, the incubation period of a certain disease in the population usually follows a certain probability distribution rather than a completely uniform distribution (Liu et al., 2020; Zhang & Wu, 2023). In contrast, distributed delays incorporate the influence of past states within a continuous time window through the integral kernel function, providing a more flexible and realistic framework for describing this heterogeneity (Frank, 2005; Smith, 2011). Unfortunately, most existing studies on distributed delays have focused on the situation in a single-patch setting (Feng et al., 2026; Ratnayake et al., 2020; Yu et al., 2019), while ignoring the interaction effects of spatial and temporal heterogeneity. To fill this gap, this paper takes into account both the cross-regional migration of individuals and the differences in their incubation periods, and establishes the following two-patch delayed differential equation system:

$$\begin{cases} \dot{S}_1(t) = A_1 - \beta_1 S_1(t) \int_0^\infty \varphi_1(\tau) I_1(t-\tau) d\tau - (\mu_1 + \alpha_{11}) S_1(t) + \alpha_{21} S_2(t), \\ \dot{I}_1(t) = \beta_1 S_1(t) \int_0^\infty \varphi_1(\tau) I_1(t-\tau) d\tau - (\mu_1 + c_1 + \alpha_{12}) I_1(t) + \alpha_{22} I_2(t), \\ \dot{S}_2(t) = A_2 - \beta_2 S_2(t) \int_0^\infty \varphi_2(\tau) I_2(t-\tau) d\tau - (\mu_2 + \alpha_{21}) S_2(t) + \alpha_{11} S_1(t), \\ \dot{I}_2(t) = \beta_2 S_2(t) \int_0^\infty \varphi_2(\tau) I_2(t-\tau) d\tau - (\mu_2 + c_2 + \alpha_{22}) I_2(t) + \alpha_{12} I_1(t), \end{cases} \quad (1.1)$$

where $S_\ell(t)$ and $I_\ell(t)$ ($\ell = 1, 2$) denote the numbers of susceptible and infected individuals in patch ℓ at time t , respectively. All the parameters in model (1.1) are nonnegative and the biological meanings are as follows: A_ℓ is the recruitment rate of susceptible individuals in patch ℓ ; β_ℓ is the transmission rate in patch ℓ ; μ_ℓ and c_ℓ are the natural death rate and the disease-induced death rate in patch ℓ , respectively; α_{11} and α_{12} are the long-time average of migration rates of susceptible and infected individuals from patch 1 to patch 2, respectively; α_{21} and α_{22} are the long-time average of migration rates of susceptible and infected individuals from patch 2 to patch 1, respectively.

Additionally, $\varphi_\ell(\tau)$ ($\ell = 1, 2$) in model (1.1) is nonnegative delay kernel function satisfying $\int_0^\infty \varphi_\ell(\tau) d\tau = 1$, which describes the distribution of the infectious delay in patch ℓ . To render the distributed delay terms in (1.1) analytically and numerically tractable, we choose $\varphi_\ell(\tau)$ ($\ell = 1, 2$) to be Erlang type, i.e.,

$$\varphi_\ell(\tau) = \frac{a_\ell^{m_\ell}}{(m_\ell - 1)!} \tau^{m_\ell - 1} e^{-a_\ell \tau}, \quad \tau \geq 0,$$

where $a_\ell > 0$ denotes the stage transition rate and $m_\ell \in \mathbb{N}$ represents the number of stages in the corresponding delay mechanism. The kernel $\varphi_\ell(\tau)$ is a probability density function, with mean delay $\mathbb{E}[\tau] = m_\ell / a_\ell$ and variance $\text{Var}(\tau) = m_\ell / a_\ell^2$. Biologically, m_ℓ quantifies the degree of heterogeneity in the delay: larger m_ℓ yields a more concentrated delay distribution (approaching a fixed delay as $m_\ell \rightarrow \infty$), whereas smaller m_ℓ produces a more dispersed waiting-time profile. It is worth noting that we chose the Erlang distribution as the kernel function for two main reasons. On the one hand, it can depict the typical epidemiological evolution processes such as incubation period, immune activation or delayed transmission in a multi-stage structure. On the other hand, the Erlang kernel can approximate any unimodal delay distribution by adjusting the shape parameter—whether highly concentrated or significantly dispersed—and the convolution term can be equivalently transformed into a finite-dimensional ordinary differential equation system by using the linear-chain technique. With this choice, the distributed delay terms in model (1.1) are written as the convolution integrals

$$X_m(t) \triangleq \int_0^\infty \varphi_1(\tau) I_1(t-\tau) d\tau, \quad Y_n(t) \triangleq \int_0^\infty \varphi_2(\tau) I_2(t-\tau) d\tau,$$

where $m = m_1$ and $n = m_2$. Defining auxiliary variables $\{X_i(t)\}_{i=1}^m$ and $\{Y_j(t)\}_{j=1}^n$, and using the linear-chain technique, we obtain the following ODE systems

$$\begin{aligned}\dot{X}_1(t) &= a_1(I_1(t) - X_1(t)), & \dot{X}_i(t) &= a_1(X_{i-1}(t) - X_i(t)), \quad i = 2, \dots, m, \\ \dot{Y}_1(t) &= a_2(I_2(t) - Y_1(t)), & \dot{Y}_j(t) &= a_2(Y_{j-1}(t) - Y_j(t)), \quad j = 2, \dots, n,\end{aligned}$$

where the chains describe the flow of individuals through m (or n) sequential latent compartments with transition rates a_1 (or a_2). The output variables $X_m(t)$ and $Y_n(t)$ coincide exactly with the original distributed delay convolutions. Consequently, the distributed delay system (1.1) can be reformulated as follows:

$$\begin{cases} \dot{S}_1(t) = A_1 - \beta_1 S_1(t) X_m(t) - (\mu_1 + \alpha_{11}) S_1(t) + \alpha_{21} S_2(t), \\ \dot{I}_1(t) = \beta_1 S_1(t) X_m(t) - (\mu_1 + c_1 + \alpha_{12}) I_1(t) + \alpha_{22} I_2(t), \\ \dot{X}_1(t) = a_1(I_1(t) - X_1(t)), \\ \dot{X}_i(t) = a_1(X_{i-1}(t) - X_i(t)), \quad i = 2, \dots, m, \\ \dot{S}_2(t) = A_2 - \beta_2 S_2(t) Y_n(t) - (\mu_2 + \alpha_{21}) S_2(t) + \alpha_{11} S_1(t), \\ \dot{I}_2(t) = \beta_2 S_2(t) Y_n(t) - (\mu_2 + c_2 + \alpha_{22}) I_2(t) + \alpha_{12} I_1(t), \\ \dot{Y}_1(t) = a_2(I_2(t) - Y_1(t)), \\ \dot{Y}_j(t) = a_2(Y_{j-1}(t) - Y_j(t)), \quad j = 2, \dots, n. \end{cases} \quad (1.2)$$

In many realistic epidemic scenarios, effective contact and migration rates are not constant over time, but are influenced by random environmental and social factors, such as seasonal variability, changes in individual behavior, mobility restrictions, and the implementation or relaxation of control measures (Zhou et al., 2023). In recent years, stochastic epidemic models have been widely used to capture such uncertainties (Kiouach & Sabbar, 2021; Liu et al., 2020, 2024; Shi et al., 2022, 2024a; Shi & Jiang, 2023; Zhang, 2025). In epidemic modeling, this randomness is typically characterized by completely memoryless white noise (Shi et al., 2022; Zhang, 2025), or by the Ornstein-Uhlenbeck (OU) process, which has memory and mean-reversion properties (Liu et al., 2024; Shi et al., 2024a; Shi & Jiang, 2023).

From an epidemiological perspective, population mobility is strongly influenced by risk perception and public health interventions. During an outbreak, individuals often reduce travel in response to increasing infection risk, media reports, or government restrictions, while mobility gradually recovers as the epidemic situation improves or control measures are relaxed (Allen, 2016). Empirical evidence from recent epidemics, particularly COVID-19, demonstrates substantial reductions in human mobility following the implementation of control measures, followed by progressive recovery once restrictions are lifted (Badr et al., 2020; Kraemer et al., 2020; Nouvellet et al., 2021). Such observations highlight the strong coupling between mobility patterns and epidemic dynamics, indicating that migration intensity does not fluctuate arbitrarily but instead oscillates around a typical baseline level determined by socioeconomic demands and habitual movement behavior (Peng et al., 2026).

The OU process provides a natural mathematical representation of this phenomenon. Its mean-reverting drift term captures the tendency of mobility to return to a long-term average level after random shocks, reflecting behavioral adaptation and policy-driven regulation. Meanwhile, the stochastic diffusion term accounts for unpredictable environmental and social disturbances. In contrast, simple white noise assumes completely uncorrelated fluctuations with no regulatory tendency, which may lead to unrealistic, persistent deviations in migration intensity. Therefore, the OU process offers a more epidemiologically grounded description of stochastic migration by incorporating both short-term variability and gradual restoration toward typical mobility levels (Han & Jiang, 2023).

In this study, we assume that the environmental noise influencing migration is governed by the following OU process

$$dx(t) = -\theta x(t) dt + \sigma dB(t),$$

where $\theta > 0$ denotes the rate of mean reversion, $\sigma > 0$ measures the intensity of the random perturbation, and $B(t)$ is a standard Brownian motion. The deterministic drift term $-\theta x(t)$ pulls the process back toward its long-term mean (taken to be zero without loss of generality), while the stochastic diffusion term $\sigma dB(t)$ introduces continuous random fluctuations. By incorporating this OU-type perturbation into the deterministic model (1.2), we arrive at the following stochastic model:

$$\begin{cases} dx(t) = -\theta x(t) dt + \sigma dB(t), \\ dS_1(t) = [A_1 - \beta_1 S_1(t) X_m(t) - (\mu_1 + \alpha_{11} e^{x(t)}) S_1(t) + \alpha_{21} e^{x(t)} S_2(t)] dt, \\ dI_1(t) = [\beta_1 S_1(t) X_m(t) - (\mu_1 + c_1 + \alpha_{12} e^{x(t)}) I_1(t) + \alpha_{22} e^{x(t)} I_2(t)] dt, \\ dX_1(t) = a_1(I_1(t) - X_1(t)) dt, \\ dX_i(t) = a_1(X_{i-1}(t) - X_i(t)) dt, \quad i = 2, \dots, m, \\ dS_2(t) = [A_2 - \beta_2 S_2(t) Y_n(t) - (\mu_2 + \alpha_{21} e^{x(t)}) S_2(t) + \alpha_{11} e^{x(t)} S_1(t)] dt, \\ dI_2(t) = [\beta_2 S_2(t) Y_n(t) - (\mu_2 + c_2 + \alpha_{22} e^{x(t)}) I_2(t) + \alpha_{12} e^{x(t)} I_1(t)] dt, \\ dY_1(t) = a_2(I_2(t) - Y_1(t)) dt, \\ dY_j(t) = a_2(Y_{j-1}(t) - Y_j(t)) dt, \quad j = 2, \dots, n. \end{cases} \quad (1.3)$$

Stochastic model (1.3) constitutes a hybrid stochastic model consisting of a stochastic differential equation for the environmental process $x(t)$ and random differential equations for the population variables, in which the OU process modulates

the migration (or movement-related) terms through the positive multiplicative factor $e^{x(t)}$. This choice guarantees positivity of the migration rates and reflects the idea that environmental perturbations can instantaneously amplify or suppress the effective coupling between the two patches. When $x(t) > 0$, migration intensities are temporarily enhanced, whereas negative values of $x(t)$ correspond to the reduced movement. Especially, the stochastic model (1.3) reduces to its deterministic counterpart (1.2) when $\sigma \rightarrow 0$.

Models (1.2) and (1.3) are two-patch epidemic models with deterministic and stochastic migration incorporating distributed delays. The migration rates between patches are modulated by an OU process, allowing us to capture random but mean-reverting environmental effects, while distributed delays are introduced through linear-chain structures to model non-instantaneous infection dynamics. This naturally arises the following questions: *For the deterministic model (1.2), can the basic reproduction number serve as a single, precise epidemiological threshold to determine disease extinction and long-term prevalence? For the corresponding stochastic model (1.3), can a new threshold criterion be constructed that is explicitly related to basic reproduction number of the deterministic model (1.2) and incorporates the intensity of random perturbations, enabling the precise identification of disease transmission risks in stochastic scenarios? Furthermore, in typical migration scenarios such as the cross-regional transportation of sheep, how to quantitatively analyze the mechanism by which migration affects the prevalence scale of brucellosis in different regions? Will the dynamic changes in the intensity of random migration reshape the cross-regional transmission pattern?*

The rest of this paper is organized as follows: In Section 2, we study the deterministic model (1.2) and establish threshold results associated with the disease-free and endemic equilibria, including asymptotic stability and persistence properties. In Section 3, we derive sufficient conditions for almost surely exponential extinction and prove the existence of a stationary distribution under a stochastic persistence threshold for the stochastic model (1.3). Section 4 provides extensive numerical experiments to validate the theoretical findings and to examine how migration intensity and OU noise parameters jointly affect the long-term infection burden and the overall epidemic scale. Section 5 concludes the paper with a discussion and possible extensions.

For convenience, we define the following notations:

$$\begin{aligned}\mathcal{Z}_1(t) &\triangleq (S_1(t), I_1(t), X_1(t), \dots, X_m(t), S_2(t), I_2(t), Y_1(t), \dots, Y_n(t)) \in \mathbb{R}_+^{m+n+4}, \\ \mathcal{Z}_2(t) &\triangleq (x(t), S_1(t), I_1(t), X_1(t), \dots, X_m(t), S_2(t), I_2(t), Y_1(t), \dots, Y_n(t)) \in \mathbb{R} \times \mathbb{R}_+^{m+n+4}, \\ \mathcal{Z}_3(t) &\triangleq (I_1(t), X_1(t), \dots, X_m(t), I_2(t), Y_1(t), \dots, Y_n(t)) \in \mathbb{R}_+^{m+n+2}.\end{aligned}$$

When no confusion arises, we abbreviate $\mathcal{Z}_1(t)$, $\mathcal{Z}_2(t)$, $\mathcal{Z}_3(t)$, $x(t)$, $S_\ell(t)$, $I_\ell(t)$, $X_i(t)$, and $Y_j(t)$ as $\mathcal{Z}_1$, $\mathcal{Z}_2$, $\mathcal{Z}_3$, x , S_ℓ , I_ℓ , X_i , and Y_j , respectively ($\ell = 1, 2, i = 1, \dots, m, j = 1, \dots, n$) in the subsequent sections.

2. Dynamics of the deterministic model (1.2)

This section is devoted to analyzing the dynamical behaviors of model (1.2).

Model (1.2) has a disease-free equilibrium $E^0 = (S_1^0, 0, \dots, 0, S_2^0, 0, \dots, 0) \in \mathbb{R}_+^{m+n+4}$, where

$$\begin{aligned}S_1^0 &= \frac{A_1 P_2 + A_2 \alpha_{21}}{P_1 P_2 - \alpha_{11} \alpha_{21}}, \quad S_2^0 = \frac{A_2 P_1 + A_1 \alpha_{11}}{P_1 P_2 - \alpha_{11} \alpha_{21}}, \\ P_1 &= \mu_1 + \alpha_{11}, \quad P_2 = \mu_2 + \alpha_{21}.\end{aligned}$$

Based on the next generation matrix method (van den Driessche & Watmough, 2002; van den Driessche, 2017), we define

$$\mathcal{F} = \begin{bmatrix} \beta_1 S_1 X_m \\ 0 \\ \vdots \\ 0 \\ \beta_2 S_2 Y_n \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathcal{V} = \begin{bmatrix} P_3 I_1 - \alpha_{22} I_2 \\ a_1 (X_1 - I_1) \\ a_1 (X_2 - X_1) \\ \vdots \\ a_1 (X_m - X_{m-1}) \\ P_4 I_2 - \alpha_{12} I_1 \\ a_2 (Y_1 - I_2) \\ a_2 (Y_2 - Y_1) \\ \vdots \\ a_2 (Y_n - Y_{n-1}) \end{bmatrix},$$

where $P_3 = \mu_1 + c_1 + \alpha_{12}$, $P_4 = \mu_2 + c_2 + \alpha_{22}$. Thus, one has

$$F = \begin{bmatrix} 0 & \cdots & 0 & \beta_1 S_1^0 & 0 & \cdots & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \beta_2 S_2^0 \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} P_3 & & & & -\alpha_{22} & 0 & \cdots & 0 \\ -a_1 & a_1 & & & 0 & 0 & \cdots & 0 \\ & \ddots & \ddots & & \vdots & \vdots & & \vdots \\ & & -a_1 & a_1 & 0 & 0 & \cdots & 0 \\ -\alpha_{12} & 0 & \cdots & 0 & P_4 & & & \\ 0 & 0 & \cdots & 0 & -a_2 & a_2 & & \\ \vdots & \vdots & & \vdots & & \ddots & \ddots & \\ 0 & 0 & \cdots & 0 & & & -a_2 & a_2 \end{bmatrix}, \quad (2.1)$$

where all omitted entries of F and V are zero. The basic reproduction number of model (1.2) is

$$\mathcal{R}_0 = \rho(FV^{-1}) = \frac{\beta_1 S_1^0 P_4 + \beta_2 S_2^0 P_3 + \sqrt{(\beta_1 S_1^0 P_4 - \beta_2 S_2^0 P_3)^2 + 4\beta_1 S_1^0 \beta_2 S_2^0 \alpha_{22} \alpha_{12}}}{2(P_3 P_4 - \alpha_{12} \alpha_{22})},$$

where $\rho(\cdot)$ represents the spectral radius of a matrix.

Lemma 2.1. *The positively invariant set of model (1.2) is*

$$\Omega = \left\{ \mathcal{Z}_1 \in \mathbb{R}_+^{m+n+4} : S_\ell + I_\ell \geq \Lambda_\ell, \ell = 1, 2, S_1 + S_2 + I_1 + I_2 \leq \bar{N}, \right. \\ \left. \Lambda_1 \leq X_i \leq \bar{N}, \Lambda_2 \leq Y_j \leq \bar{N}, i = 1, \dots, m, j = 1, \dots, n \right\},$$

where

$$\Lambda_1 = \frac{A_1 Q_2 + A_2 \hat{\alpha}_2}{Q_1 Q_2 - \hat{\alpha}_1 \hat{\alpha}_2}, \quad \Lambda_2 = \frac{A_2 Q_1 + A_1 \hat{\alpha}_1}{Q_1 Q_2 - \hat{\alpha}_1 \hat{\alpha}_2}, \quad \bar{N} = \frac{A_1 + A_2}{\underline{\mu}}, \\ Q_1 = \mu_1 + c_1 + \check{\alpha}_1, \quad Q_2 = \mu_2 + c_2 + \check{\alpha}_2, \quad \underline{\mu} = \min\{\mu_1, \mu_2\} \\ \hat{\alpha}_1 = \min\{\alpha_{11}, \alpha_{12}\}, \quad \check{\alpha}_1 = \max\{\alpha_{11}, \alpha_{12}\}, \quad \hat{\alpha}_2 = \min\{\alpha_{21}, \alpha_{22}\}, \quad \check{\alpha}_2 = \max\{\alpha_{21}, \alpha_{22}\}.$$

Proof. Let $N_1 = S_1 + I_1$ and $N_2 = S_2 + I_2$. It then follows from (1.2) that

$$\begin{cases} \dot{N}_1 \geq A_1 - Q_1 N_1 + \hat{\alpha}_2 N_2, \\ \dot{N}_2 \geq A_2 - Q_2 N_2 + \hat{\alpha}_1 N_1. \end{cases}$$

Construct the following auxiliary system

$$\begin{cases} \dot{x}_1 = A_1 - Q_1 x_1 + \hat{\alpha}_2 x_2, \\ \dot{x}_2 = A_2 - Q_2 x_2 + \hat{\alpha}_1 x_1. \end{cases} \quad (2.2)$$

Obviously, (Λ_1, Λ_2) is the unique equilibrium of linear system (2.2), and its characteristic equation at (Λ_1, Λ_2) is

$$\lambda^2 + (Q_1 + Q_2)\lambda + Q_1 Q_2 - \hat{\alpha}_1 \hat{\alpha}_2 = 0. \quad (2.3)$$

All roots of equation (2.3) have negative real parts since $Q_1 + Q_2 > 0$ and $Q_1 Q_2 - \hat{\alpha}_1 \hat{\alpha}_2 > 0$, which means that (Λ_1, Λ_2) is globally asymptotically stable, and thus

$$\lim_{t \rightarrow +\infty} (x_1(t), x_2(t)) = (\Lambda_1, \Lambda_2).$$

Based on the comparison principle, $\liminf_{t \rightarrow +\infty} N_\ell(t) \geq \Lambda_\ell$ for $\ell = 1, 2$. Furthermore, let $N = N_1 + N_2$. We can obtain from (1.2) that

$$\begin{aligned} \dot{N} &\leq A_1 + A_2 - \mu_1 S_1 - \mu_1 I_1 - \mu_2 S_2 - \mu_2 I_2 \\ &\leq A_1 + A_2 - \underline{\mu} N, \end{aligned}$$

from which one has $\limsup_{t \rightarrow +\infty} N(t) \leq \bar{N}$. Similarly, we have

$$\begin{aligned}\Lambda_1 &\leq \liminf_{t \rightarrow +\infty} X_i(t) \leq \limsup_{t \rightarrow +\infty} X_i(t) \leq \bar{N}, \quad i = 1, \dots, m, \\ \Lambda_2 &\leq \liminf_{t \rightarrow +\infty} Y_j(t) \leq \limsup_{t \rightarrow +\infty} Y_j(t) \leq \bar{N}, \quad j = 1, \dots, n.\end{aligned}$$

Thus, the positively invariant set of model (1.2) is Ω . The proof is completed.

Theorem 2.1. The disease-free equilibrium E^0 of model (1.2) is locally asymptotically stable if $\mathcal{R}_0 < 1$, and unstable if $\mathcal{R}_0 > 1$.

Proof. By changing the order of the state variables in model (1.2) to $(S_1, S_2, I_1, X_1, \dots, X_m, I_2, Y_1, \dots, Y_n)$, we can calculate the Jacobian matrix of the model at disease-free equilibrium E^0 as

$$J = \begin{bmatrix} J_1 & * \\ O & F - V \end{bmatrix},$$

where “*” denotes $2 \times (m + n + 2)$ non-zero matrix, F and V are defined in (2.1), and

$$J_1 = \begin{bmatrix} -(\mu_1 + \alpha_{11}) & \alpha_{21} \\ \alpha_{11} & -(\mu_2 + \alpha_{21}) \end{bmatrix}.$$

To determine the signs of the real parts of the eigenvalues of matrix J , it suffices to consider those of matrices J_1 and $F - V$. The characteristic equation corresponding to matrix J_1 is

$$\lambda^2 + (\mu_1 + \mu_2 + \alpha_{11} + \alpha_{21})\lambda + (\mu_1 + \alpha_{11})(\mu_2 + \alpha_{21}) - \alpha_{11}\alpha_{21} = 0.$$

It then follows from $\mu_1 + \mu_2 + \alpha_{11} + \alpha_{21} > 0$ and $(\mu_1 + \alpha_{11})(\mu_2 + \alpha_{21}) - \alpha_{11}\alpha_{21} > 0$ that both eigenvalues of J_1 have negative real parts. For matrix $F - V$, we define

$$\lambda_{\max}(F - V) \triangleq \max\{\operatorname{Re} z : z \text{ is an eigenvalue of } F - V\}. \quad (2.4)$$

Then, from [36, Theorem 2], one has

$$\mathcal{R}_0 > 1 \Leftrightarrow \lambda_{\max}(F - V) > 0, \quad \mathcal{R}_0 < 1 \Leftrightarrow \lambda_{\max}(F - V) < 0. \quad (2.5)$$

Therefore, all the eigenvalues of J have negative real parts if $\mathcal{R}_0 < 1$, which means the disease-free equilibrium E^0 of model (1.2) is locally asymptotically stable. Conversely, matrix J has at least one eigenvalue with a positive real part if $\mathcal{R}_0 > 1$, so E^0 is unstable.

Theorem 2.2. The disease-free equilibrium E^0 of model (1.2) is globally asymptotically stable if $\mathcal{R}_0 < 1$.

Proof. From Lemma 2.1 and Theorem 2.1, we only need to prove that E^0 is globally attractive. It follows from model (1.2) that

$$\begin{cases} \dot{S}_1 \leq A_1 - P_1 S_1 + \alpha_{21} S_2, \\ \dot{S}_2 \leq A_2 - P_2 S_2 + \alpha_{11} S_1. \end{cases}$$

Consider the following auxiliary model

$$\begin{cases} \dot{x}_1 = A_1 - P_1 x_1 + \alpha_{21} x_2, \\ \dot{x}_2 = A_2 - P_2 x_2 + \alpha_{11} x_1. \end{cases} \quad (2.6)$$

Similar to the deduction in model (2.2), we can conclude that the unique positive equilibrium (S_1^0, S_2^0) of model (2.6) is locally asymptotically stable. Further, since (2.6) is a linear model, (S_1^0, S_2^0) is globally asymptotically stable, which means that for any $\varpi > 0$, there exists a $t_1 > 0$ such that $S_\ell(t) \leq S_\ell^0 + \varpi$ for $t \geq t_1$ and $\ell = 1, 2$. Therefore, for $t \geq t_1$, we have

$$\begin{cases} \dot{I}_1 \leq \beta_1(S_1^0 + \varpi)X_m - (\mu_1 + c_1 + \alpha_{12})I_1 + \alpha_{22}I_2, \\ \dot{X}_1 = a_1(I_1 - X_1), \\ \dot{X}_i = a_1(X_{i-1} - X_i), \quad i = 2, \dots, m, \\ \dot{I}_2 \leq \beta_2(S_2^0 + \varpi)Y_n - (\mu_2 + c_2 + \alpha_{22})I_2 + \alpha_{12}I_1, \\ \dot{Y}_1 = a_2(I_2 - Y_1), \\ \dot{Y}_j = a_2(Y_{j-1} - Y_j), \quad j = 2, \dots, n. \end{cases}$$

Construct the auxiliary model $\dot{\tilde{z}}_1 = \mathcal{J}_1 \tilde{z}_1$ with $\tilde{z}_1 = (y_1, X_1, \dots, X_m, y_2, Y_1, \dots, Y_n)^\top$, $\mathcal{J}_1 = F - V + \varpi \mathcal{J}^*$, and

$$\mathcal{J}^* = \begin{bmatrix} 0 & \cdots & 0 & \beta_1 & 0 & \cdots & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \beta_2 \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}. \quad (2.7)$$

Clearly, $\dot{\tilde{z}}_1 = \mathcal{J}_1 \tilde{z}_1$ has unique equilibrium $(0, \dots, 0) \in \mathbb{R}^{m+n+2}$ because the matrix $\mathcal{J}_1$ is invertible. From (2.5), $\lambda_{\max}(F - V) < 0$ when $\mathcal{R}_0 < 1$. Thus, we can select $\varpi > 0$ sufficiently small such that $\lambda_{\max}(F - V + \varpi \mathcal{J}^*) < 0$, which implies that all the eigenvalues of $\mathcal{J}_1$ have negative real parts, and then $\tilde{z}_1 \rightarrow 0(t \rightarrow +\infty)$. Based on the comparison principle, we have $I_\ell(t) \rightarrow 0$, $X_i(t) \rightarrow 0$ and $Y_j(t) \rightarrow 0$ as $t \rightarrow +\infty$ ($\ell = 1, 2$, $i = 1, \dots, m$, $j = 1, \dots, n$).

The following limiting system

$$\begin{cases} \dot{S}_1 = A_1 - P_1 S_1 + \alpha_{21} S_2, \\ \dot{S}_2 = A_2 - P_2 S_2 + \alpha_{11} S_1, \end{cases}$$

has unique positive equilibrium (S_1^0, S_2^0) , which is globally asymptotically stable. Therefore, we can conclude that the disease-free equilibrium E^0 is globally asymptotically stable according to the theory of asymptotic autonomous systems (Thieme, 1992).

Theorem 2.3. When $\mathcal{R}_0 > 1$, model (1.2) has at least one endemic equilibrium $E^* = (S_1^*, I_1^*, X_1^*, \dots, X_m^*, S_2^*, I_2^*, Y_1^*, \dots, Y_n^*)$, and there exists a positive constant ε such that each solution $\mathcal{Z}_1(t)$ of model (1.2) for arbitrary initial value $\mathcal{Z}_1(0) \in \mathbb{R}_+^{m+n+4}$ with $I_1(0) + I_2(0) > 0$ satisfies

$$\liminf_{t \rightarrow +\infty} S_\ell(t) \geq \varepsilon, \quad \liminf_{t \rightarrow +\infty} I_\ell(t) \geq \varepsilon, \quad \ell = 1, 2, \\ \liminf_{t \rightarrow +\infty} X_i(t) \geq \varepsilon, \quad \liminf_{t \rightarrow +\infty} Y_j(t) \geq \varepsilon, \quad i = 1, \dots, m, j = 1, \dots, n.$$

Proof. From Lemma 2.1, there exists $T_1 > 0$ such that $X_m(t) \leq \bar{N} + \varsigma$ and $Y_n(t) \leq \bar{N} + \varsigma$ for $t > T_1$ and any given $\varsigma > 0$. Thus, we have

$$\dot{S}_\ell \geq A_\ell - [\beta_\ell(\bar{N} + \varsigma) + \mu_\ell + \alpha_{\ell 1}] S_\ell, \quad t > T_1, \ell = 1, 2,$$

which means

$$\liminf_{t \rightarrow +\infty} S_\ell(t) \geq \frac{A_\ell}{2[\beta_\ell(\bar{N} + \varsigma) + \mu_\ell + \alpha_{\ell 1}]}.$$

According to (1.2), if we can show that $I_\ell(t)$ ($\ell = 1, 2$) is permanent, then $X_i(t)$ and $Y_j(t)$ ($i = 1, \dots, m$, $j = 1, \dots, n$) are also permanent, and thus the disease is permanent. Therefore, we just need to show $\liminf_{t \rightarrow +\infty} I_\ell(t) \geq \varepsilon$ ($\ell = 1, 2$) in the following.

Let

$$\begin{aligned} \Gamma &= \{ \mathcal{Z}_1 | S_\ell \geq 0, I_\ell \geq 0, X_i \geq 0, Y_j \geq 0, \ell = 1, 2, i = 1, \dots, m, j = 1, \dots, n \}, \\ \Gamma_0 &= \{ \mathcal{Z}_1 \in \Gamma | I_1 + I_2 + X_1 + \dots + X_m + Y_1 + \dots + Y_n > 0 \}, \\ \partial\Gamma_0 &= \Gamma \setminus \Gamma_0. \end{aligned}$$

Clearly, Γ is positively invariant, and $\partial\Gamma_0$ is relatively closed in Γ . Thus, it is sufficient to show that $I_\ell(t)$ ($\ell = 1, 2$) is uniformly persistent with respect to $(\Gamma_0, \partial\Gamma_0)$, which will be proven by the following two steps.

Step 1: Define $\Gamma_\partial = \{ \mathcal{Z}_1(0) | \mathcal{Z}_1(t) \in \partial\Gamma_0, \forall t \geq 0 \}$. We will show that $\mathcal{W} \triangleq \bigcup \{ \omega(\mathcal{Z}_1(0)) | \mathcal{Z}_1(0) \in \Gamma_\partial \}$ is isolated and acyclic, and E^0 is a cover of $\mathcal{W}$, where $\omega(\mathcal{Z}_1(0))$ denotes the ω -limit set of the solution of model (1.2) with initial condition $\mathcal{Z}_1(0) \in \Gamma$. For $\mathcal{Z}_1(0) \in \Gamma_\partial$, if there exists a $t_0 > 0$ such that $X_1(t_0) > 0, \dots, X_m(t_0) > 0, Y_1(t_0) > 0, \dots, Y_n(t_0) > 0, I_1(t_0) > 0$, or $I_2(t_0) > 0$, then $\mathcal{Z}_1(t_0) \in \Gamma_0$, which contradicts $\mathcal{Z}_1(0) \in \Gamma_\partial$. Therefore, we have

$$\Gamma_\partial = \{(S_1, 0, \dots, 0, S_2, 0, \dots, 0) \in \mathbb{R}^{m+n+4} | S_\ell \geq 0, \ell = 1, 2\}.$$

For any $Z_1(0) \in \Gamma_\partial$, we have $I_\ell(t) = 0$, $X_i(t) = 0$, and $Y_j(t) = 0$ ($\ell = 1, 2, i = 1, \dots, m, j = 1, \dots, n$). Thus, $S_\ell(t)$ ($\ell = 1, 2$) satisfies

$$\begin{cases} \dot{S}_1 = A_1 - (\mu_1 + \alpha_{11})S_1 + \alpha_{21}S_2, \\ \dot{S}_2 = A_2 - (\mu_2 + \alpha_{21})S_2 + \alpha_{11}S_1, \end{cases}$$

which implies that $\lim_{t \rightarrow +\infty} S_\ell(t) = S_\ell^0$ ($\ell = 1, 2$). Therefore, $\mathcal{W} = \{E^0\}$, which means that $\mathcal{W}$ is isolated and acyclic, and E^0 covers $\mathcal{W}$.

Step 2: We prove

$$W^s(E^0) \cap \Gamma_0 = \emptyset, \quad (2.8)$$

by contradiction, where $W^s(E^0)$ represents the stable manifold of E^0 . If (2.8) does not hold, we can find a solution $Z(t) \in \Gamma_0$ ($t \geq 0$) of model (1.2) starting in $Z(0) \in \Gamma_0$ such that

$$\lim_{t \rightarrow +\infty} S_\ell(t) = S_\ell^0, \lim_{t \rightarrow +\infty} I_\ell(t) = 0, \lim_{t \rightarrow +\infty} X_i(t) = 0, \lim_{t \rightarrow +\infty} Y_j(t) = 0, \ell = 1, 2, i = 1, \dots, m, j = 1, \dots, n, \quad (2.9)$$

which means for any $\delta > 0$ ($\delta < \min\{S_1^0, S_2^0\}$), there exists a $T_2 > T_1 > 0$ such that

$$|S_\ell(t) - S_\ell^0| < \delta, 0 < I_\ell(t) < \delta, 0 < X_i(t) < \delta, 0 < Y_j(t) < \delta, \quad t \geq T_2, \ell = 1, 2, i = 1, \dots, m, j = 1, \dots, n.$$

Therefore, when $t \geq T_2$, we can obtain from (1.2) that

$$\begin{cases} \dot{S}_1 \geq A_1 - \tilde{P}_1 S_1 + \alpha_{21} S_2, \\ \dot{S}_2 \geq A_2 - \tilde{P}_2 S_2 + \alpha_{11} S_1, \end{cases} \quad (2.10)$$

where $\tilde{P}_1 = \beta_1 \delta + \mu_1 + \alpha_{11}$, $\tilde{P}_2 = \beta_2 \delta + \mu_2 + \alpha_{21}$. The auxiliary system

$$\begin{cases} \dot{\mathcal{S}}_1 = A_1 - \tilde{P}_1 \mathcal{S}_1 + \alpha_{21} \mathcal{S}_2, \\ \dot{\mathcal{S}}_2 = A_2 - \tilde{P}_2 \mathcal{S}_2 + \alpha_{11} \mathcal{S}_1. \end{cases} \quad (2.11)$$

has a unique positive equilibrium $(\mathcal{S}_1^*, \mathcal{S}_2^*) = \left(\frac{A_1 \tilde{P}_2 + A_2 \alpha_{21}}{P_1 P_2 - \alpha_{11} \alpha_{21}}, \frac{A_2 \tilde{P}_1 + A_1 \alpha_{11}}{P_1 P_2 - \alpha_{11} \alpha_{21}} \right)$, and $\lim_{\delta \rightarrow 0} (\mathcal{S}_1^*, \mathcal{S}_2^*) = (S_1^0, S_2^0)$. Since (2.11) is a linear

system and all the eigenvalues of the Jacobian matrix at $(\mathcal{S}_1^*, \mathcal{S}_2^*)$ have negative real parts, the equilibrium $(\mathcal{S}_1^*, \mathcal{S}_2^*)$ is globally asymptotically stable. Meanwhile, it follows from (2.4) and (2.5) that $\lambda_{\max}(F - V) > 0$ for $\mathcal{R}_0 > 1$, which means we can choose $\varrho > 0$ ($\varrho < \min\{S_1^0, S_2^0\}$) sufficiently small such that $\lambda_{\max}(F - V - \varrho \mathcal{J}^*) > 0$. Therefore, for the above given $\varrho > 0$, we can find $\delta > 0$ sufficiently small such that $\mathcal{S}_\ell^* \geq S_\ell^0 - \varrho$ ($\ell = 1, 2$). Based on the comparison principle, there exists a $T_3 > T_2 > 0$ such that $S_\ell(t) \geq S_\ell^0 - \varrho$ ($\ell = 1, 2$) for $t \geq T_3$. It then follows from (1.2) that

$$\begin{cases} \dot{I}_1 \geq \beta_1 (S_1^0 - \varrho) X_m - (\mu_1 + c_1 + \alpha_{12}) I_1 + \alpha_{22} I_2, \\ \dot{X}_1 = a_1 (I_1 - X_1), \\ \dot{X}_i = a_1 (X_{i-1} - X_i), \quad i = 2, \dots, m, \\ \dot{I}_2 \geq \beta_2 (S_2^0 - \varrho) Y_n - (\mu_2 + c_2 + \alpha_{22}) I_2 + \alpha_{12} I_1, \\ \dot{Y}_1 = a_2 (I_2 - Y_1), \\ \dot{Y}_j = a_2 (Y_{j-1} - Y_j), \quad j = 2, \dots, n, \end{cases}$$

for $t \geq T_3$.

Construct the auxiliary system $\dot{\tilde{Z}}_2 = (F - V - \varrho \mathcal{J}^*) \tilde{Z}_2$, where $\tilde{Z}_2 = (\tilde{I}_1, X_1, \dots, X_m, \tilde{I}_2, Y_1, \dots, Y_m)^\top$, and $\mathcal{J}^*$ is defined in (2.7). Since $\mathcal{R}_0 > 1$, we have $\lambda_{\max}(F - V - \varrho \mathcal{J}^*) > 0$ for the above chosen $\varrho > 0$. Hence, we have $\lim_{t \rightarrow +\infty} I_\ell(t) = \infty$, $\lim_{t \rightarrow +\infty} X_i(t) = \infty$ and $\lim_{t \rightarrow +\infty} Y_j(t) = \infty$ ($\ell = 1, 2, i = 2, \dots, m, j = 2, \dots, n$), which contradicts the results in (2.9), and thus (2.8) holds. Therefore,

we can conclude that model (1.2) is uniformly persistent with respect to $(\Gamma_0, \partial\Gamma_0)$ and has at least one endemic equilibrium E^* based on (Thieme, 1993 Theorem 4.6) and (Hutson & Schmitt, 1992). This completes the proof.

3. Dynamics of the stochastic model (1.3)

In this section, we study the existence and uniqueness of the global solution, stationary distribution and the extinction of disease for stochastic model (1.3).

3.1. Existence and uniqueness of the global solution

Theorem 3.1. For any initial value $Z_2(0) \in \mathbb{R} \times \mathbb{R}_+^{m+n+4}$, there exists a unique solution $Z_2(t)$ of the model (1.3) for $t \geq 0$ and this solution will remain in $\mathbb{R} \times \mathbb{R}_+^{m+n+4}$ almost surely (a.s.).

proof. The proof is standard and follows from the classical localization argument together with the construction of a suitable Lyapunov function to preclude finite-time explosion and ensure positivity. For brevity, we omit the routine preliminary steps and only present the key Lyapunov function used in the argument. Let

$$V_1 = (S_1 - 1 - \ln S_1) + (I_1 - 1 - \ln I_1) + \frac{\beta_1}{a_1} \sum_{i=1}^m (X_i - 1 - \ln X_i) + (S_2 - 1 - \ln S_2) + (I_2 - 1 - \ln I_2) \\ + \frac{\beta_2}{a_2} \sum_{j=1}^n (Y_j - 1 - \ln Y_j) + (e^x - 1 - x).$$

Then, we have

$$\begin{aligned} \mathcal{L}V_1 &= \left(1 - \frac{1}{S_1}\right) [A_1 - \beta_1 S_1 X_m - (\mu_1 + \alpha_{11} e^x) S_1 + \alpha_{21} e^x S_2] \\ &\quad + \left(1 - \frac{1}{I_1}\right) [\beta_1 S_1 X_m - (\mu_1 + c_1 + \alpha_{12} e^x) I_1 + \alpha_{22} e^x I_2] \\ &\quad + \beta_1 \left(1 - \frac{1}{X_1}\right) (I_1 - X_1) + \beta_1 \sum_{i=2}^m \left(1 - \frac{1}{X_i}\right) (X_{i-1} - X_i) \\ &\quad + \left(1 - \frac{1}{S_2}\right) [A_2 - \beta_2 S_2 Y_n - (\mu_2 + \alpha_{21} e^x) S_2 + \alpha_{11} e^x S_1] \\ &\quad + \left(1 - \frac{1}{I_2}\right) [\beta_2 S_2 Y_n - (\mu_2 + c_2 + \alpha_{22} e^x) I_2 + \alpha_{12} e^x I_1] \\ &\quad + \beta_2 \left(1 - \frac{1}{Y_1}\right) (I_2 - Y_1) + \beta_2 \sum_{j=2}^n \left(1 - \frac{1}{Y_j}\right) (Y_{j-1} - Y_j) + \theta x(1 - e^x) + \frac{\sigma^2 e^x}{2} \\ &\leq A_1 + A_2 + 2\mu_1 + c_1 + m\beta_1 + 2\mu_2 + c_2 + n\beta_2 + f_1(x) + \beta_1 I_1 + \beta_2 I_2, \end{aligned}$$

where $f_1(x) = (\alpha_{11} + \alpha_{12} + \alpha_{21} + \alpha_{22} + \frac{\sigma^2}{2})e^x + \theta x(1 - e^x)$. Note that $\lim_{x \rightarrow \pm\infty} f_1(x) = -\infty$, hence $f_1(x)$ has an upper limit. Set $N(t) \triangleq S_1(t) + I_1(t) + S_2(t) + I_2(t)$, a direct computation yields

$$dN(t) = \sum_{k=1}^2 (A_k - \mu_k(S_k(t) + I_k(t)) - c_k I_k(t)) dt \leq (A_1 + A_2) - \underline{\mu} N(t) dt,$$

which implies that $N(t) \leq \bar{N}$ if $N(0) \leq \bar{N}$. Using $I_1(t), I_2(t) \leq \max\{\bar{N}, I_1(0), I_2(0)\}$, we can easily obtain

$$\mathcal{L}V_1 \leq A_1 + A_2 + (2\mu_1 + c_1) + m\beta_1 + (2\mu_2 + c_2) + n\beta_2 + \sup_{x \in \mathbb{R}} f_1(x) + \max\{\beta_1, \beta_2\} \max\{\bar{N}, I_1(0), I_2(0)\} = M_1,$$

where $M_1 < \infty$ is a positive constant. The remaining steps are standard and can be found, for instance, in (Khasminskii, 2012; Mao, 2007). This completes the proof.

Remark 3.1. From the stochastic model (1.3), one has

$$\dot{N}(t) = \sum_{k=1}^2 (A_k - \mu_k(S_k(t) + I_k(t)) - c_k I_k(t)) \leq A_1 + A_2 - \underline{\mu} N(t).$$

Whenever $N(0) \leq \bar{N}$, we have

$$N(t) \leq \bar{N}, \quad t \geq 0.$$

Hence, the set

$$\bar{\Omega} \triangleq \left\{ \mathcal{Z}_2(t) \in \mathbb{R} \times \mathbb{R}_+^{m+n+4} : N(t) \leq \bar{N} \right\}$$

is positively invariant for (1.3). In particular, for any solution with initial condition in $\bar{\Omega}$,

$$0 \leq S_1(t), I_1(t), S_2(t), I_2(t) \leq \bar{N}, \quad t \geq 0, \text{ a.s.}$$

Throughout the rest of the paper, we will always assume that the initial condition is taken in the positively invariant set $\bar{\Omega}$.

The following lemma recalls a standard ergodic property of the OU process, which plays a key role in our analysis.

Lemma 3.1. ((Shi et al., 2024a)) For the stochastic differential equation

$$dx(t) = -\theta x(t) dt + \sigma dB(t),$$

where $\theta > 0$ and $\sigma > 0$ are constants and $B(t)$ is a standard Brownian motion, one has

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t e^{x(\tau)} d\tau = e^{\frac{\sigma^2}{4\theta}}, \quad \text{a.s.}$$

Proof. Observe that

$$\frac{1}{t} \int_0^t e^{x(\tau)} d\tau = \frac{1}{t} \int_0^t e^{\frac{\sqrt{2\theta}}{\sigma} x(\tau) \cdot \frac{\sigma}{\sqrt{2\theta}}} d\tau.$$

Let

$$v(t) = \frac{\sqrt{2\theta}}{\sigma} x(t).$$

Then $v(t)$ is a stationary Gaussian process with standard normal distribution. By the ergodic property of the OU process, we obtain

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t e^{x(\tau)} d\tau = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} e^{\frac{\sigma}{\sqrt{2\theta}} v} dv.$$

Completing the square gives

$$\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(v - \frac{\sigma}{\sqrt{2\theta}}\right)^2}{2}} e^{\frac{\sigma^2}{4\theta}} dv = e^{\frac{\sigma^2}{4\theta}},$$

which completes the proof.

3.2. Extinction

In this subsection, we investigate conditions under which the infection becomes extinct in the presence of environmental stochasticity. Unlike the deterministic case, where extinction is typically characterized by the global stability of the disease-free equilibrium, random perturbations may significantly alter the long-term behavior of the system. Our aim here is to derive a sufficient condition under which the infected components decay to zero almost surely in the presence of OU-driven environmental fluctuations. By constructing suitable Lyapunov functions and exploiting the structural properties of the model, we establish explicit threshold-type criteria under which the disease cannot persist and eventually dies out.

Lemma 3.2. ((Farina & Rinaldi, 2000)) Based on the stochastic model (1.3), we construct the following system

$$\dot{\mathcal{Z}}_3(t) = A(x(t), S_1(t), S_2(t))\mathcal{Z}_3(t),$$

where $A(\cdot)$ is Metzler. Let $V(\mathcal{Z}_3) \triangleq \|\mathcal{Z}_3\|_1$. Then

$$\dot{V}(\mathcal{Z}_3(t)) \leq \rho_1(S_1(t), S_2(t))V(\mathcal{Z}_3(t)), \quad (3.1)$$

where

$$\rho_1(S_1, S_2) = \max\{0, a_1 - (\mu_1 + c_1), \beta_1 S_1 - a_1, a_2 - (\mu_2 + c_2), \beta_2 S_2 - a_2\}. \quad (3.2)$$

Proof. Since A is Metzler and $V(\mathcal{Z}_3) = \|\mathcal{Z}_3\|_1$, the standard matrix-measure inequality gives

$$\dot{V}(\mathcal{Z}_3) \leq \mu^{(1)}(A) V(\mathcal{Z}_3),$$

where $\mu^{(1)}(\cdot)$ denotes the matrix measure induced by $\|\cdot\|_1$. For Metzler matrices, $\mu^{(1)}(A)$ equals the maximum column sum. Let $c(A) \triangleq \mathbf{1}^\top A = (c_1(A), c_2(A), \dots, c_{m+n+2}(A))$ be the column-sum vector with $c_j(A) = \sum_i a_{ij}$. A direct identification of $A(x, S_1, S_2)$ from (1.3) yields

$$c(A(x, S_1, S_2)) = \left(a_1 - (\mu_1 + c_1), \underbrace{0, \dots, 0}_{m-1}, \beta_1 S_1 - a_1, a_2 - (\mu_2 + c_2), \underbrace{0, \dots, 0}_{n-1}, \beta_2 S_2 - a_2 \right).$$

Consequently,

$$\mu^{(1)}(A(x, S_1, S_2)) = \max_{1 \leq j \leq m+n+2} c_j(A(x, S_1, S_2)) = \max\{0, a_1 - (\mu_1 + c_1), \beta_1 S_1 - a_1, a_2 - (\mu_2 + c_2), \beta_2 S_2 - a_2\}.$$

This is exactly (3.2) and hence implies (3.1). The proof is completed.

Theorem 3.2. Assume the initial condition belongs to $\bar{\Omega}$ so that $S_1(t), S_2(t) \leq \bar{N}$ for all $t \geq 0$. Define

$$\mathcal{R}_0^e = \max\left\{\frac{a_1}{\mu_1 + c_1}, \frac{a_2}{\mu_2 + c_2}, \frac{\beta_1 \bar{N}}{a_1}, \frac{\beta_2 \bar{N}}{a_2}\right\}.$$

If $\mathcal{R}_0^e < 1$, then there exist $\gamma > 0$ and a constant $C > 0$ such that

$$V(\mathcal{Z}_3(t)) \leq Ce^{-\gamma t} V(\mathcal{Z}_3(0)), \quad t \geq 0, \text{ a.s.}$$

Consequently,

$$I_1(t), I_2(t), X_i(t) \ (1 \leq i \leq m), Y_j(t) \ (1 \leq j \leq n) \rightarrow 0(t \rightarrow +\infty) \text{ a.s.}$$

Proof. Choose $\eta > 0$ such that

$$a_1 \leq (\mu_1 + c_1) - \eta, \quad a_2 \leq (\mu_2 + c_2) - \eta, \quad \beta_1 \bar{N} \leq a_1 - \eta, \quad \beta_2 \bar{N} \leq a_2 - \eta.$$

Since $S_1(t), S_2(t) \leq \bar{N}$ for all $t \geq 0$,

$$\beta_1 S_1(t) - a_1 \leq \beta_1 \bar{N} - a_1 \leq -\eta, \quad \beta_2 S_2(t) - a_2 \leq -\eta, \quad \forall t \geq 0.$$

Together with $a_1 - (\mu_1 + c_1) \leq -\eta$ and $a_2 - (\mu_2 + c_2) \leq -\eta$, Lemma 3.2 and (3.2) imply

$$\dot{V}(\mathcal{Z}_3(t)) \leq -\eta V(\mathcal{Z}_3(t)), \quad t \geq 0, \text{ a.s.}$$

Integrating yields $V(\mathcal{Z}_3(t)) \leq e^{-\eta t} V(\mathcal{Z}_3(0))$ for all $t \geq 0$. Taking $\gamma \triangleq \eta$ and $C \triangleq 1$ finishes the proof.

3.3. Stationary distribution

In deterministic epidemic models, long-term persistence is usually reflected by the existence of a positive endemic equilibrium: when the basic reproduction number is larger than one, the infection does not die out and the solution typically approaches an interior steady state, so that the disease remains at a constant endemic level. Once environmental fluctuations are incorporated, such pointwise convergence is generally no longer expected, because the random perturbations continuously disturb the system. In this situation, persistence should be understood in a probabilistic sense. A common and practical way to express stochastic persistence is the existence of a stationary distribution for the model: the state variables do not drift to extinction, but instead exhibit long-run stable fluctuations, leading to a statistically steady endemic pattern. Hence, proving the existence of a stationary distribution provides a rigorous mathematical description of disease persistence under random environmental effects.

Define stochastic threshold of model (1.3) as

$$\mathcal{R}_0^s = \frac{\beta_2 \bar{S}_2 + \frac{\alpha_{12} \alpha_{22}}{\mu_1 + c_1 + \alpha_{12} e^{\frac{\sigma^2}{4\theta}} - \beta_1 \bar{S}_1}}{\mu_2 + c_2 + \alpha_{22} e^{\frac{\sigma^2}{4\theta}}},$$

where

$$\bar{S}_1 = \frac{A_1 \mu_2 + (A_1 + A_2) \alpha_{21} e^{\frac{\sigma^2}{4\theta}}}{\mu_1 \mu_2 + (\mu_1 \alpha_{21} + \mu_2 \alpha_{11}) e^{\frac{\sigma^2}{4\theta}}}, \quad \bar{S}_2 = \frac{A_2 \mu_1 + (A_1 + A_2) \alpha_{11} e^{\frac{\sigma^2}{4\theta}}}{\mu_1 \mu_2 + (\mu_1 \alpha_{21} + \mu_2 \alpha_{11}) e^{\frac{\sigma^2}{4\theta}}}.$$

Theorem 3.3. *The stochastic model (1.3) admits a stationary distribution when $\mathcal{R}_0^s > 1$.*

proof. The proof proceeds in three main steps. First, we construct appropriate stochastic Lyapunov functions for the Markov process generated by (1.3) and derive the corresponding drift inequalities. Second, based on these Lyapunov estimates, we identify a suitable compact set that the process returns to with positive frequency. Third, combining the drift condition with the recurrence property on the compact set, we verify the required ergodic stability, which yields the existence of a stationary distribution.

Let

$$\tilde{S}_1 = \frac{S_1}{\bar{S}_1}, \quad \tilde{S}_2 = \frac{S_2}{\bar{S}_2}.$$

Applying the Itô's formula to $-\ln S_1$, we have

$$\begin{aligned} \mathcal{L}(-\ln S_1) &= -\frac{A_1}{\bar{S}_1} - \frac{\alpha_{21} e^x S_2}{\bar{S}_1} + \beta_1 X_m + \mu_1 + \alpha_{11} e^x = \\ &= -\frac{A_1}{\bar{S}_1 \bar{S}_1} - \frac{\alpha_{21} e^x \bar{S}_2 \tilde{S}_2}{\bar{S}_1 \bar{S}_1} + \beta_1 X_m + \mu_1 + \alpha_{11} e^{\frac{\sigma^2}{4\theta}} + \alpha_{11} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right) - \frac{A_1}{\bar{S}_1} \left(\frac{1}{\bar{S}_1} - 1 \right) - \frac{\alpha_{21} \bar{S}_2}{\bar{S}_1} \left(\frac{e^x \tilde{S}_2}{\bar{S}_1} - 1 \right) \\ &\quad + \beta_1 X_m + \alpha_{11} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right), \end{aligned}$$

where the last equation holds due to

$$\begin{cases} A_1 + \alpha_{21} e^{\frac{\sigma^2}{4\theta}} \bar{S}_2 = \left(\mu_1 + \alpha_{11} e^{\frac{\sigma^2}{4\theta}} \right) \bar{S}_1, \\ A_2 + \alpha_{11} e^{\frac{\sigma^2}{4\theta}} \bar{S}_1 = \left(\mu_2 + \alpha_{21} e^{\frac{\sigma^2}{4\theta}} \right) \bar{S}_2. \end{cases}$$

According to the inequality $z - 1 \geq \ln z$ for all $z > 0$, it follows that

$$\begin{aligned}\mathcal{L}(-\ln S_1) &\leq -\frac{A_1}{\bar{S}_1} \ln \frac{1}{\bar{S}_1} - \frac{\alpha_{21} \bar{S}_2}{\bar{S}_1} \ln \frac{e^x \tilde{S}_2}{\bar{S}_1} + \beta_1 X_m + \alpha_{11} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right) \\ &= \frac{A_1 + \alpha_{21} \bar{S}_2}{\bar{S}_1} \ln \tilde{S}_1 - \frac{\alpha_{21} \bar{S}_2}{\bar{S}_1} (\ln \tilde{S}_2 + x) + \beta_1 X_m + \alpha_{11} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right).\end{aligned}$$

Using a similar inequality analysis as above, one obtains

$$\mathcal{L}(-\ln S_2) \leq \frac{A_2 + \alpha_{11} \bar{S}_1}{\bar{S}_2} \ln \tilde{S}_2 - \frac{\alpha_{11} \bar{S}_1}{\bar{S}_2} \ln(\tilde{S}_1 + x) + \beta_2 Y_n + \alpha_{21} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right).$$

Let $\bar{I}_2 = \bar{Y}_n = 1$ and

$$\bar{I}_1 = \bar{X}_m = \frac{\alpha_{22}}{\mu_1 + c_1 + \alpha_{12} e^{\frac{\sigma^2}{4\theta}} - \beta_1 \bar{S}_1}, \quad \tilde{I}_1 = \frac{I_1}{\bar{I}_1}, \quad \tilde{I}_2 = \frac{I_2}{\bar{I}_2}, \quad \tilde{X}_m = \frac{X_m}{\bar{X}_m}, \quad \tilde{Y}_n = \frac{Y_n}{\bar{Y}_n}.$$

Then we compute

$$\begin{aligned}\mathcal{L}(-\ln I_1) &= -\frac{\beta_1 \bar{S}_1 X_m + \alpha_{22} e^x I_2}{I_1} + \mu_1 + c_1 + \alpha_{12} e^{\frac{\sigma^2}{4\theta}} + \alpha_{12} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right) = \\ &\leq -\frac{\beta_1 \bar{S}_1 \bar{X}_m}{\bar{I}_1} \left(\frac{\tilde{S}_1 \tilde{X}_m}{\tilde{I}_1} - 1 \right) - \frac{\alpha_{22} \bar{I}_2}{\bar{I}_1} \left(\frac{e^x \tilde{I}_2}{\tilde{I}_1} - 1 \right) \\ &\quad + \alpha_{12} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right) \frac{\beta_1 \bar{S}_1 \bar{X}_m + \alpha_{22} \bar{I}_2}{\bar{I}_1} \ln \tilde{I}_1 - \frac{\beta_1 \bar{S}_1 \bar{X}_m}{\bar{I}_1} (\ln \tilde{S}_1 + \ln \tilde{X}_m) - \frac{\alpha_{22} \bar{I}_2}{\bar{I}_1} (\ln \tilde{I}_2 + x) + \alpha_{12} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right).\end{aligned}$$

In addition, one gets

$$\begin{aligned}\mathcal{L} \left(-\sum_{i=1}^m \ln X_i \right) &= a_1 \left(1 - \frac{I_1}{X_1} \right) + a_1 \left(1 - \frac{X_1}{X_2} \right) + \dots + a_1 \left(1 - \frac{X_{m-1}}{X_m} \right) \leq \\ &= -a_1 \ln \frac{I_1}{X_1} - a_1 \ln \frac{X_1}{X_2} - \dots - a_1 \ln \frac{X_{m-1}}{X_m} a_1 (\ln X_m - I_1) = a_1 (\ln \tilde{X}_m - \tilde{I}_1).\end{aligned}$$

Similarly, one has

$$\mathcal{L} \left(-\sum_{j=1}^n \ln Y_j \right) \leq a_2 (\ln \tilde{Y}_n - \ln \tilde{I}_2).$$

Furthermore, we have

$$\begin{aligned}
\mathcal{L}(-\ln I_2) &= -\frac{\beta_2 \bar{S}_2 \bar{Y}_n + \alpha_{12} e^x \bar{I}_1}{\bar{I}_2} + \mu_2 + c_2 + \alpha_{22} e^{\frac{\sigma^2}{4\theta}} + \alpha_{22} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right) = \\
&\leq -\frac{\beta_2 \bar{S}_2 \bar{Y}_n}{\bar{I}_2} \left(\frac{\tilde{S}_2 \tilde{Y}_n}{\tilde{I}_2} - 1 \right) - \frac{\alpha_{12} \bar{I}_1}{\bar{I}_2} \left(\frac{e^x \tilde{I}_1}{\tilde{I}_2} - 1 \right) + \alpha_{22} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right) + \mu_2 + c_2 + \alpha_{22} e^{\frac{\sigma^2}{4\theta}} \\
&\quad - \frac{\beta_2 \bar{S}_2 \bar{Y}_n + \alpha_{12} \bar{I}_1}{\bar{I}_2} \frac{\beta_2 \bar{S}_2 \bar{Y}_n + \alpha_{12} \bar{I}_1}{\bar{I}_2} \ln \tilde{I}_2 - \frac{\beta_2 \bar{S}_2 \bar{Y}_n}{\bar{I}_2} (\ln \tilde{S}_2 + \ln \tilde{Y}_n) - \frac{\alpha_{12} \bar{I}_1}{\bar{I}_2} (\ln \tilde{I}_1 + x) \\
&\quad + \alpha_{22} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right) + \mu_2 + c_2 + \alpha_{22} e^{\frac{\sigma^2}{4\theta}} - \frac{\beta_2 \bar{S}_2 \bar{Y}_n + \alpha_{12} \bar{I}_1}{\bar{I}_2} = -(\mathcal{R}_0^s - 1) \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{\sigma^2}{4\theta}} \right) \\
&\quad + \frac{\beta_2 \bar{S}_2 \bar{Y}_n + \alpha_{12} \bar{I}_1}{\bar{I}_2} \ln \tilde{I}_2 - \frac{\beta_2 \bar{S}_2 \bar{Y}_n}{\bar{I}_2} (\ln \tilde{S}_2 + \ln \tilde{Y}_n) - \frac{\alpha_{12} \bar{I}_1}{\bar{I}_2} (\ln \tilde{I}_1 + x) + \alpha_{22} \left(e^x - e^{\frac{\sigma^2}{4\theta}} \right),
\end{aligned}$$

where

$$\mathcal{R}_0^s = \frac{\beta_2 \bar{S}_2 + \alpha_{12} \bar{I}_1}{\mu_2 + c_2 + \alpha_{22} e^{\frac{\sigma^2}{4\theta}}} = \frac{\beta_2 \bar{S}_2 + \frac{\alpha_{12} \alpha_{22}}{\sigma^2}}{\mu_1 + c_1 + \alpha_{12} e^{\frac{\sigma^2}{4\theta}} - \beta_1 \bar{S}_1}.$$

Then define

$$V_2 = -\ln I_2 - p_1 \ln S_1 - p_2 \ln S_2 - p_3 \ln I_1 - p_4 \sum_{i=1}^m \ln X_i - p_5 \sum_{j=1}^n \ln Y_j - \frac{p_6}{\theta} x,$$

where

$$p_1 = \frac{\bar{S}_1^2 (A_2 \alpha_{12} \beta_1 \bar{I}_1^2 + \bar{S}_1 \alpha_{11} \alpha_{12} \beta_1 \bar{I}_1^2 + \bar{S}_2 \alpha_{22}^2 \beta_2)}{\alpha_{22} (A_1 A_2 + A_1 \bar{S}_1 \alpha_{11} + A_2 \bar{S}_2 \alpha_{12} + \bar{S}_1 \bar{S}_2 \alpha_{11} \alpha_{12} - \bar{S}_1 \bar{S}_2 \alpha_{12} \alpha_{22})},$$

$$p_2 = \frac{\bar{S}_2^2 (A_1 \alpha_{22} \beta_2 + \bar{S}_1 \alpha_{12}^2 \beta_1 \bar{I}_1^2 + \bar{S}_2 \alpha_{12} \alpha_{22} \beta_2)}{\alpha_{22} (A_1 A_2 + A_1 \bar{S}_1 \alpha_{11} + A_2 \bar{S}_2 \alpha_{12} + \bar{S}_1 \bar{S}_2 \alpha_{11} \alpha_{12} - \bar{S}_1 \bar{S}_2 \alpha_{12} \alpha_{22})},$$

$$p_3 = \frac{\alpha_{12} \bar{I}_1^2}{\alpha_{22}}, \quad p_4 = \frac{\bar{S}_1 \beta_1}{a_1} p_3, \quad p_5 = \frac{\beta_2 \bar{S}_2}{a_2}, \quad p_6 = \frac{\alpha_{12} \bar{I}_1}{\bar{I}_2} + p_1 \frac{\alpha_{21} \bar{S}_2}{\bar{S}_1} + p_2 \frac{\alpha_{11} \bar{S}_1}{\bar{S}_2} + p_3 \frac{\alpha_{22} \bar{I}_2}{\bar{I}_1}$$

such that

$$\begin{cases} p_1 \frac{A_1 + \alpha_{21} \bar{S}_2}{\bar{S}_1} = p_2 \frac{\alpha_{22} \bar{S}_1}{\bar{S}_2} + p_3 \frac{\beta_1 \bar{S}_1 \bar{X}_m}{\bar{I}_1}, \\ p_2 \frac{A_2 + \alpha_{11} \bar{S}_1}{\bar{S}_2} = p_1 \frac{\alpha_{21} \bar{S}_2}{\bar{S}_1} + \frac{\beta_2 \bar{S}_2 \bar{Y}_n}{\bar{I}_2}, \\ p_3 \frac{\beta_1 \bar{S}_1 \bar{X}_m + \alpha_{22} \bar{I}_2}{\bar{I}_1} = p_4 a_1 + \frac{\alpha_{12} \bar{I}_1}{\bar{I}_2}, \\ \frac{\beta_2 \bar{S}_2 \bar{Y}_n + \alpha_{12} \bar{I}_1}{\bar{I}_2} = p_5 a_2 + p_3 \frac{\alpha_{22} \bar{I}_2}{\bar{I}_1}, \\ p_3 \frac{\beta_1 \bar{S}_1 \bar{X}_m}{\bar{I}_1} = p_4 a_1, \\ \frac{\beta_2 \bar{S}_2 \bar{Y}_n}{\bar{I}_2} = p_5 a_2. \end{cases}$$

Hence we have

$$\mathcal{LV}_2 \leq -(\mathcal{R}_0^S - 1) \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{\sigma^2}{4\theta}} \right) + p_1 \beta_1 X_m + p_2 \beta_2 Y_n + f_2(x),$$

where $f_2(x) = (\alpha_{22} + p_1 \alpha_{11} + p_2 \alpha_{21} + p_3 \alpha_{12})(e^x - e^{\frac{\sigma^2}{4\theta}})$.

Define

$$V_3 = V_2 + \frac{p_1 \beta_1}{a_1} \sum_{i=1}^m X_i + \frac{p_2 \beta_2}{a_2} \sum_{j=1}^n Y_j - \frac{p_1 \beta_1 e^{\kappa_1}}{\alpha_{12}} I_2,$$

where κ_1 is a positive constant to be determined later. Then one gets

$$\mathcal{LV}_3 \leq -(\mathcal{R}_0^S - 1) \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{\sigma^2}{4\theta}} \right) + p_1 \beta_1 \left(1 - e^{x + \frac{1}{\kappa_1}} \right) I_1 + \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 + f_2(x).$$

Define

$$\begin{aligned} V_4 = & e^x - x - 1 - \ln S_1 - \ln I_1 - \sum_{i=1}^m \frac{\ln X_i}{a_1} - \ln S_2 - \sum_{j=1}^n \frac{\ln Y_j}{a_2} - \ln(\bar{N} - S_1 - I_1 - S_2 - I_2) \\ & + X_1^{p_7+1} + (X_1 + X_2)^{p_7+1} + \dots + \left(\sum_{i=1}^m X_m \right)^{p_7+1} + Y_1^{p_7+1} + (Y_1 + Y_2)^{p_7+1} + \dots + \left(\sum_{j=1}^n Y_j \right)^{p_7+1}, \end{aligned}$$

where $p_7 \in (0, 1)$ is a constant. By model (1.3) and the fact that X_i, Y_j are driven linearly by I_1, I_2 , one gets

$$0 \leq X_i(t) \leq \bar{N} \quad (i = 1, \dots, m), \quad 0 \leq Y_j(t) \leq \bar{N} \quad (j = 1, \dots, n), \quad t \geq 0, \text{ a.s.}$$

Therefore,

$$\begin{aligned}
\mathcal{LV}_4 \leq & \theta x(1 - e^x) + \frac{\sigma^2 e^x}{2} - \frac{A_1}{S_1} + \beta_1 X_m + \mu_1 + \alpha_{11} e^x - \frac{\alpha_{22} e^x I_2}{I_1} + \mu_1 + c_1 + \alpha_{12} e^x - \frac{I_1}{X_1} - \sum_{i=2}^m \frac{X_{i-1}}{X_i} + m \\
& - \frac{A_2}{S_2} + \beta_2 Y_n + \mu_2 + \alpha_{21} e^x - \frac{I_2}{Y_1} - \sum_{j=2}^n \frac{Y_{j-1}}{Y_j} + n + \frac{A_1 + A_2 - \mu(S_1 + I_1 + S_2 + I_2) - c_2 I_2}{\bar{N} - S_1 - I_1 - S_2 - I_2} \\
& + p_7 a_1 \left[(\bar{N} - X_1) X_1^{p_7} + (\bar{N} - X_2) X_2^{p_7} + \dots + (\bar{N} - X_m) X_m^{p_7} \right] \\
& + p_7 a_2 \left[(\bar{N} - Y_1) Y_1^{p_7} + (\bar{N} - Y_2) Y_2^{p_7} + \dots + (\bar{N} - Y_n) Y_n^{p_7} \right] \\
\leq & \theta x(1 - e^x) + \frac{\sigma^2 e^x}{2} - \frac{A_1}{S_1} - \frac{\alpha_{22} e^x I_2}{I_1} - \frac{I_1}{X_1} - \sum_{i=2}^m \frac{X_{i-1}}{X_i} - \frac{A_2}{S_2} - \frac{I_2}{Y_1} - \sum_{j=2}^n \frac{Y_{j-1}}{Y_j} \\
& - \frac{c_2 I_2}{\bar{N} - S_1 - I_1 - S_2 - I_2} - \frac{p_7 a_1}{2} \sum_{i=1}^m X_i^{p_7+1} - \frac{p_7 a_2}{2} \sum_{j=1}^n Y_j^{p_7+1} \\
& + 2\mu_1 + \underline{\mu} + c_1 + m + n + (\beta_1 + \beta_2) \bar{N} + M_1 + (\alpha_{11} + \alpha_{12} + \alpha_{21}) e^{\frac{\sigma^2}{4\theta}} + f_3(x),
\end{aligned}$$

where

$$\begin{aligned}
M_1 = & \sup_{(X_1, \dots, X_m, Y_1, \dots, Y_n) \in \mathbb{R}_+^{m+n}} \left\{ p_7 a_1 \left[(\bar{N} - X_1/2) X_1^{p_7} + (\bar{N} - X_2/2) X_2^{p_7} + \dots + (\bar{N} - X_m/2) X_m^{p_7} \right] \right. \\
& \left. + p_7 a_2 \left[(\bar{N} - Y_1/2) Y_1^{p_7} + (\bar{N} - Y_2/2) Y_2^{p_7} + \dots + (\bar{N} - Y_n/2) Y_n^{p_7} \right] \right\}
\end{aligned}$$

and $f_3(x) = (\alpha_{11} + \alpha_{12} + \alpha_{21})(e^x - e^{\frac{\sigma^2}{4\theta}})$.

Define

$$V_5 = M_2 V_3 + V_4,$$

where M_2 is a sufficiently large constant satisfying

$$\begin{aligned}
& -M_2(\mathcal{R}_0^s - 1) \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{\sigma^2}{4\theta}} \right) + \sup_{x \in \mathbb{R}} \left\{ \theta x(1 - e^x) + \frac{\sigma^2 e^x}{2} \right\} \\
& + 2\mu_1 + \underline{\mu} + c_1 + m + n + (\beta_1 + \beta_2) \bar{N} + M_1 + (\alpha_{11} + \alpha_{12} + \alpha_{21}) e^{\frac{\sigma^2}{4\theta}} \leq -2.
\end{aligned} \tag{3.3}$$

Then we have

$$\mathcal{LV}_5 \leq f_4(\mathcal{Z}_2) + M_2 f_2(x) + f_3(x),$$

with

$$\begin{aligned}
f_4 = & -M_2(\mathcal{R}_0^s - 1) \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{\sigma^2}{4\theta}} \right) + M_2 p_1 \beta_1 \left(1 - e^{x + \frac{1}{\kappa_1}} \right) I_1 + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 \\
& + \theta x(1 - e^x) + \frac{\sigma^2 e^x}{2} - \frac{A_1}{S_1} - \frac{\alpha_{22} e^x I_2}{I_1} - \frac{I_1}{X_1} - \sum_{i=2}^m \frac{X_{i-1}}{X_i} - \frac{A_2}{S_2} - \frac{I_2}{Y_1} - \sum_{j=2}^n \frac{Y_{j-1}}{Y_j} \\
& - \frac{c_2 I_2}{\bar{N} - S_1 - I_1 - S_2 - I_2} - \frac{p_7 a_1}{2} \sum_{i=1}^m X_i^{p_7+1} - \frac{p_7 a_2}{2} \sum_{j=1}^n Y_j^{p_7+1} \\
& + 2\mu_1 + \underline{\mu} + c_1 + m + n + (\beta_1 + \beta_2) \bar{N} + M_1 + (\alpha_{11} + \alpha_{12} + \alpha_{21}) e^{\frac{\sigma^2}{4\theta}}.
\end{aligned} \tag{3.4}$$

Construct a compact set

$$\mathbb{D} = \left\{ \mathcal{Z}_2 \in \bar{\Omega} : |x| \leq \frac{1}{\kappa_1}, \kappa_2 \leq S_1, \kappa_2^2 \leq I_1, \kappa_2^{j+2} \leq X_i \leq \frac{1}{\kappa_2}, \right. \\ \left. \kappa_2 \leq S_2, \kappa_2 \leq I_2, \kappa_2^{j+1} \leq Y_j \leq \frac{1}{\kappa_2}, S_1 + I_1 + S_2 + I_2 \leq \bar{N} - \kappa_2^2 \right\}$$

such that $f_4(\mathcal{Z}_2) \leq -1$ for any $\mathcal{Z}_2 \in \bar{\Omega} \setminus \mathbb{D} \triangleq \mathbb{D}^c = \bigcup_{i=1}^{7+2m+2n} \mathbb{D}_i^c$, where

$$\begin{aligned} \mathbb{D}_1^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x > \frac{1}{\kappa_1} \right\}, & \mathbb{D}_2^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x < -\frac{1}{\kappa_1} \right\}, \\ \mathbb{D}_3^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x \leq \frac{1}{\kappa_1}, I_2 < \kappa_2 \right\}, & \mathbb{D}_4^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x \leq \frac{1}{\kappa_1}, S_1 < \kappa_2 \right\}, \\ \mathbb{D}_5^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : -\frac{1}{\kappa_1} \leq x \leq \frac{1}{\kappa_1}, I_2 \geq \kappa_2, I_1 < \kappa_2^2 \right\}, & \mathbb{D}_6^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x \leq \frac{1}{\kappa_1}, S_2 < \kappa_2 \right\}, \\ \mathbb{D}_7^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x \leq \frac{1}{\kappa_1}, I_2 \geq \kappa_2, \bar{N} - (S_1 + I_1 + S_2 + I_2) < \kappa_2^2 \right\}, \\ \mathbb{D}_8^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x \leq \frac{1}{\kappa_1}, I_1 \geq \kappa_2^2, X_1 < \kappa_2^3 \right\}, \\ \mathbb{D}_9^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x \leq \frac{1}{\kappa_1}, I_2 \geq \kappa_2, Y_1 < \kappa_2^2 \right\}, \\ \mathbb{D}_{8+t_1}^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x \leq \frac{1}{\kappa_1}, X_{t_1-1} \geq \kappa_2^{t_1+1}, X_{t_1} < \kappa_2^{t_1+2} \right\}, \quad t_1 = 2, \dots, m, \\ \mathbb{D}_{7+m+t_2}^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x \leq \frac{1}{\kappa_1}, Y_{t_2-1} \geq \kappa_2^{t_2}, Y_{t_2} < \kappa_2^{t_2+1} \right\}, \quad t_2 = 2, \dots, n, \\ \mathbb{D}_{7+m+n+t_3}^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x \leq \frac{1}{\kappa_1}, X_{t_3} > \frac{1}{\kappa_2} \right\}, \quad t_3 = 1, \dots, m, \\ \mathbb{D}_{7+2m+n+t_4}^c &= \left\{ \mathcal{Z}_2 \in \bar{\Omega} : x \leq \frac{1}{\kappa_1}, Y_{t_4} > \frac{1}{\kappa_2} \right\}, \quad t_4 = 1, \dots, n. \end{aligned}$$

Let κ_1 and κ_2 be two small enough positive constants satisfying

$$\begin{aligned} \min \left\{ -\frac{\theta}{2\kappa_1} e^{\frac{1}{\kappa_1}}, -\frac{\theta}{2\kappa_1} \left(1 - e^{-\frac{1}{\kappa_1}} \right) \right\} + \sup_{x \in \mathbb{R}} \left\{ -\frac{\theta x}{2} (e^x - 1) + \frac{\sigma^2}{2} e^x + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \alpha_{22} M_2 \bar{N} e^x}{\alpha_{12}} \right\} \\ + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2)}{\alpha_{12}} \right) \bar{N} + 2\mu_1 + \underline{\mu} + c_1 \\ + m + n + (\beta_1 + \beta_2) \bar{N} + M_1 + (\alpha_{11} + \alpha_{12} + \alpha_{21}) e^{\frac{\sigma^2}{4\theta}} \leq -1, \end{aligned} \quad (3.5)$$

$$M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \kappa_2 \leq 1, \quad (3.6)$$

$$\min \left\{ -\frac{A_1}{\kappa_2}, -\frac{\alpha_{22}e^{-\frac{1}{\kappa_1}}}{\kappa_2}, -\frac{1}{\kappa_2}, -\frac{A_2}{\kappa_2}, -\frac{c_2}{\kappa_2}, -\frac{p_7a_1}{2\kappa_2^{p_7+1}}, -\frac{p_7a_2}{2\kappa_2^{p_7+1}} \right\} \\ + M_2 \left(p_2\beta_2 + \frac{p_1\beta_1e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22}e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} \leq 1. \quad (3.7)$$

Based on the different inequalities used in the proof, we will prove $f_4(\mathcal{Z}_2) \leq -1$ in the following three cases.

Case I. For $\mathcal{Z}_2 \in \mathbb{D}_1^c$ or $\mathcal{Z}_2 \in \mathbb{D}_2^c$, i.e., $|x|$ is sufficiently large, we prove $f_4(\mathcal{Z}_2) \leq -1$ by using (3.5).

(i) If $\mathcal{Z}_2 \in \mathbb{D}_1^c$, from (3.4) and (3.5), we have

$$f_4 \leq \theta x(1 - e^x) + \frac{\sigma^2}{2}e^x + M_2 \left(p_2\beta_2 + \frac{p_1\beta_1e^{\frac{1}{\kappa_1}}(\mu_2 + c_2 + \alpha_{22}e^x)}{\alpha_{12}} \right) I_2 \\ + 2\mu_1 + \underline{\mu} + c_1 + m + n + (\beta_1 + \beta_2)\bar{N} + M_1 + (\alpha_{11} + \alpha_{12} + \alpha_{21})e^{\frac{\sigma^2}{4\theta}} \\ \leq -\frac{\theta}{2\kappa_1}e^{\frac{1}{\kappa_1}} + \sup_{x \in \mathbb{R}} \left\{ -\frac{\theta x}{2}(e^x - 1) + \frac{\sigma^2}{2}e^x + \frac{p_1\beta_1e^{\frac{1}{\kappa_1}}\alpha_{22}M_2\bar{N}e^x}{\alpha_{12}} \right\} \\ + M_2 \left(p_2\beta_2 + \frac{p_1\beta_1e^{\frac{1}{\kappa_1}}(\mu_2 + c_2)}{\alpha_{12}} \right) \bar{N} + 2\mu_1 + \underline{\mu} + c_1 + m + n \\ + (\beta_1 + \beta_2)\bar{N} + M_1 + (\alpha_{11} + \alpha_{12} + \alpha_{21})e^{\frac{\sigma^2}{4\theta}} \\ \leq -1.$$

(ii) If $\mathcal{Z}_2 \in \mathbb{D}_2^c$, combining (3.4) and (3.5), we obtain

$$\begin{aligned}
f_4 &\leq \theta x(1 - e^x) + \frac{\sigma^2}{2}e^x + M_2 \left(p_2\beta_2 + \frac{p_1\beta_1 e^{\frac{1}{\kappa_1}}(\mu_2 + c_2 + \alpha_{22}e^x)}{\alpha_{12}} \right) I_2 \\
&\quad + 2\mu_1 + \underline{\mu} + c_1 + m + n + (\beta_1 + \beta_2)\bar{N} + M_1 + (\alpha_{11} + \alpha_{12} + \alpha_{21})e^{\frac{\sigma^2}{4\theta}} \\
&\leq -\frac{\theta}{2\kappa_1} \left(1 - e^{-\frac{1}{\kappa_1}} \right) + \sup_{x \in \mathbb{R}} \left\{ -\frac{\theta x}{2}(e^x - 1) + \frac{\sigma^2}{2}e^x + \frac{p_1\beta_1 e^{\frac{1}{\kappa_1}}\alpha_{22}M_2\bar{N}e^x}{\alpha_{12}} \right\} \\
&\quad + M_2 \left(p_2\beta_2 + \frac{p_1\beta_1 e^{\frac{1}{\kappa_1}}(\mu_2 + c_2)}{\alpha_{12}} \right) \bar{N} + 2\mu_1 + \underline{\mu} + c_1 + m + n \\
&\quad + (\beta_1 + \beta_2)\bar{N} + M_1 + (\alpha_{11} + \alpha_{12} + \alpha_{21})e^{\frac{\sigma^2}{4\theta}} \\
&\leq -1.
\end{aligned}$$

Case II. For $\mathcal{Z}_2 \in \mathbb{D}_3^c$, i.e., I_2 is sufficiently small, by using (3.3), (3.4) and (3.6), it follows that

$$\begin{aligned}
f_4 &\leq M_2 \left(p_2\beta_2 + \frac{p_1\beta_1 e^{\frac{1}{\kappa_1}}(\mu_2 + c_2 + \alpha_{22}e^x)}{\alpha_{12}} \right) I_2 - 2 \\
&\leq M_2 \left(p_2\beta_2 + \frac{p_1\beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22}e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \kappa_2 - 2 \\
&\leq -1.
\end{aligned}$$

Case III. For $\mathcal{Z}_2 \in \mathbb{D}_i^c$ ($i = 4, \dots, 7 + 2m + 2n$), i.e., other state variables tends to the boundary of $\bar{\Omega}$, we prove $f_4(\mathcal{Z}_2) \leq -1$ by using (3.7).

(i) If $\mathcal{Z}_2 \in \mathbb{D}_4^c$, by virtue of (3.3), (3.4) and (3.7), we have

$$\begin{aligned}
f_4 &\leq -\frac{A_1}{S_1} + M_2 \left(p_2\beta_2 + \frac{p_1\beta_1 e^{\frac{1}{\kappa_1}}(\mu_2 + c_2 + \alpha_{22}e^x)}{\alpha_{12}} \right) I_2 - 2 \leq -\frac{A_1}{\kappa_2} + M_2 \left(p_2\beta_2 + \frac{p_1\beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22}e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} - 2 \\
&\leq -1.
\end{aligned}$$

(ii) If $\mathcal{Z}_2 \in \mathbb{D}_5^c$, in view of (3.3), (3.4) and (3.7), it is immediate that

$$\begin{aligned} f_4 &\leq -\frac{\alpha_{22}e^x I_2}{I_1} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 - 2 \\ &\leq -\frac{\alpha_{22}e^{\frac{1}{\kappa_1}}}{\kappa_2} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} - 2 \leq -1. \end{aligned}$$

(iii) If $\mathcal{Z}_2 \in \mathbb{D}_6^c$, it follows from (3.3), (3.4) and (3.7) that

$$\begin{aligned} f_4 &\leq -\frac{A_2}{S_2} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 - 2 \leq -\frac{A_2}{\kappa_2} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} - 2 \\ &\leq -1. \end{aligned}$$

(iv) If $\mathcal{Z}_2 \in \mathbb{D}_7^c$, employing (3.3), (3.4) and (3.7) yields

$$\begin{aligned} f_4 &\leq -\frac{c_2 I_2}{\bar{N} - S_1 - I_1 - S_2 - I_2} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 - 2 \\ &\leq -\frac{c_2}{\kappa_2} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} - 2 \leq -1. \end{aligned}$$

(v) If

 $\mathcal{Z}_2 \in \mathbb{D}_8^\zeta$, with (3.3), (3.4) and (3.7) at hand, we deduce

$$f_4 \leq -\frac{I_1}{X_1} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 - 2 \leq -\frac{1}{\kappa_2} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} - 2$$

$$\leq -1.$$

(vi) If $\mathcal{Z}_2 \in \mathbb{D}_9^\zeta$, collecting (3.3), (3.4) and (3.7) gives

$$f_4 \leq -\frac{I_2}{Y_1} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 - 2 \leq -\frac{1}{\kappa_2} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} - 2$$

$$\leq -1.$$

(vii) If $\mathcal{Z}_2 \in \mathbb{D}_{8+\iota_1}^\zeta$, $\iota_1 = 2, \dots, m$, in view of (3.3), (3.4) and (3.7), it is immediate that

$$f_4 \leq -\sum_{i=2}^m \frac{X_{i-1}}{X_i} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 - 2$$

$$\leq -\frac{1}{\kappa_2} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} - 2 \leq -1.$$

(viii) If $\mathcal{Z}_2 \in \mathbb{D}_{7+m+\iota_2}^c$, $\iota_2 = 2, \dots, n$, from (3.3), (3.4) and (3.7), we obtain

$$\begin{aligned} f_4 &\leq -\sum_{j=2}^n \frac{Y_{j-1}}{Y_j} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 - 2 \\ &\leq -\frac{1}{\kappa_2} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} - 2 \leq -1. \end{aligned}$$

(ix) If $\mathcal{Z}_2 \in \mathbb{D}_{7+m+n+\iota_3}^c$, $\iota_3 = 1, \dots, m$, it follows from (3.3), (3.4) and (3.7) that

$$\begin{aligned} f_4 &\leq -\frac{p_7 a_1}{2} \sum_{i=1}^m X_i^{p_7+1} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 - 2 \\ &\leq -\frac{p_7 a_1}{2 \kappa_2^{p_7+1}} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} - 2 \leq -1. \end{aligned}$$

(x) If $\mathcal{Z}_2 \in \mathbb{D}_{7+2m+n+\iota_4}^c$, $\iota_4 = 1, \dots, n$, collecting (3.3), (3.4) and (3.7) gives

$$\begin{aligned} f_4 &\leq -\frac{p_7 a_2}{2} \sum_{j=1}^n Y_j^{p_7+1} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} (\mu_2 + c_2 + \alpha_{22} e^x)}{\alpha_{12}} \right) I_2 - 2 \\ &\leq -\frac{p_7 a_2}{2 \kappa_2^{p_7+1}} + M_2 \left(p_2 \beta_2 + \frac{p_1 \beta_1 e^{\frac{1}{\kappa_1}} \left(\mu_2 + c_2 + \alpha_{22} e^{\frac{1}{\kappa_1}} \right)}{\alpha_{12}} \right) \bar{N} - 2 \leq -1. \end{aligned}$$

In a word, we arrive that $f_4(\mathcal{Z}_2) \leq -1$ for any $\mathcal{Z}_2 \in \mathbb{D}^c$.

In addition, there exists a point $(\underline{x}, \underline{S}_1, \underline{I}_1, \underline{X}_1, \dots, \underline{X}_m, \underline{S}_2, \underline{I}_2, \underline{Y}_1, \dots, \underline{Y}_n)$ in the interior of $\bar{\Omega}$ such that $f_4(\underline{x}, \underline{S}_1, \underline{I}_1, \underline{X}_1, \dots, \underline{X}_m, \underline{S}_2, \underline{I}_2, \underline{Y}_1, \dots, \underline{Y}_n)$ takes the minimum value. Thus

$$V_6(\mathcal{Z}_2) = V_5(\mathcal{Z}_2) - V_5(\underline{x}, \underline{S}_1, \underline{I}_1, \underline{X}_1, \dots, \underline{X}_m, \underline{S}_2, \underline{I}_2, \underline{Y}_1, \dots, \underline{Y}_n)$$

is a non-negative $\mathcal{C}^2$ -function. Applying the Itô's formula to V_6 , we obtain

$$\mathcal{L}V_6(\mathcal{Z}_2) \leq f_4(\mathcal{Z}_2) + M_2 f_2(x) + f_3(x).$$

For any initial value $\mathcal{Z}_2(0) \in \overline{\Omega}$ and the time interval $[0, t]$, we have

$$\begin{aligned} 0 \leq \frac{\mathbb{E}V_6(\mathcal{Z}_2(t))}{t} &\leq \frac{\mathbb{E}V_6(\mathcal{Z}_2(0))}{t} + \frac{1}{t} \int_0^t \mathbb{E}(f_4(\mathcal{Z}_2(\tau))) d\tau \\ &+ [M_2(\alpha_{22} + p_1\alpha_{11} + p_2\alpha_{21} + p_3\alpha_{12}) + (\alpha_{11} + \alpha_{12} + \alpha_{21})] \left(\mathbb{E} \left(\frac{1}{t} \int_0^t e^{x(\tau)} d\tau \right) - e^{\frac{\sigma^2}{4\theta}} \right). \end{aligned} \quad (3.8)$$

Since $x(t)$ is an ergodic OU process with stationary distribution, [Lemma 3.1](#) implies that

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t e^{x(\tau)} d\tau = e^{\frac{\sigma^2}{4\theta}}. \quad (3.9)$$

Allowing $t \rightarrow \infty$ and substituting (3.9) into (3.8), one gets

$$0 \leq \liminf_{t \rightarrow \infty} \frac{1}{t} \int_0^t \mathbb{E}(f_4(\mathcal{Z}_2(\tau))) d\tau, \quad a.s..$$

On the other hand, we note that

$$f_4(\mathcal{Z}_2) \leq M_3 < +\infty, \quad \forall \mathcal{Z}_2 \in \overline{\Omega}.$$

Then we have

$$\begin{aligned} \liminf_{t \rightarrow \infty} \frac{1}{t} \int_0^t \mathbb{E}(f_4(\mathcal{Z}_2(\tau))) d\tau &\leq M_3 \liminf_{t \rightarrow \infty} \frac{1}{t} \int_0^t \mathbf{1}_{\{\mathcal{Z}_2(\tau) \in \mathbb{D}\}} d\tau - \liminf_{t \rightarrow \infty} \frac{1}{t} \int_0^t \mathbf{1}_{\{\mathcal{Z}_2(\tau) \in \mathbb{D}^c\}} d\tau \\ &\leq (M_3 + 1) \liminf_{t \rightarrow \infty} \frac{1}{t} \int_0^t \mathbf{1}_{\{\mathcal{Z}_2(\tau) \in \mathbb{D}\}} d\tau - 1. \end{aligned}$$

Therefore, we obtain

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \int_0^t \mathbf{1}_{\{\mathcal{Z}_2(\tau) \in \mathbb{D}\}} d\tau \geq \frac{1}{M_3 + 1} > 0, \quad a.s.. \quad (3.10)$$

That is, along almost every sample path, the process spends a positive proportion of time in the compact set $\mathbb{D}$. Following arguments analogous to those used in the study of stability and ergodicity for Markov processes (see, e.g., ([Meyn & Tweedie, 2012](#))), this pathwise recurrence property implies a corresponding lower bound for the averaged transition probabilities:

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \int_0^t \mathbb{P}(\tau, \mathcal{Z}_2(\tau), \mathbb{D}) d\tau \geq \frac{1}{M_3 + 1} > 0, \quad a.s., \quad (3.11)$$

where $\mathbb{P}(t, z, \mathbb{D})$ denotes the transition probability that the solution of system (1.3), starting from state z at time 0, enters the set $\mathbb{D}$ at time t . Inequalities (3.10) and (3.11) show that the process is recurrent with respect to the compact set $\mathbb{D}$ and visits $\mathbb{D}$ with positive frequency. Consequently, the stochastic model (1.3) admits at least one stationary distribution. This completes the proof.

Remark 3.2. When noise intensity σ tends to zero, the stochastic threshold $\mathcal{R}_0^s$ will converge to basic reproduction number $\mathcal{R}_0$. This is because if $\mu_1 + c_1 + \alpha_{12}e^{\frac{\sigma^2}{4\theta}} - \beta_1\bar{S}_1 > 0$, the condition $\mathcal{R}_0^s > 1$ is equivalent to the algebraic inequality

$$\beta_1\bar{S}_1 \left(\mu_2 + c_2 + \alpha_{22}e^{\frac{\sigma^2}{4\theta}} \right) + \beta_2\bar{S}_2 \left(\mu_1 + c_1 + \alpha_{12}e^{\frac{\sigma^2}{4\theta}} \right) + \alpha_{12}\alpha_{22} > \beta_1\beta_2\bar{S}_1\bar{S}_2 + \left(\mu_1 + c_1 + \alpha_{12}e^{\frac{\sigma^2}{4\theta}} \right) \left(\mu_2 + c_2 + \alpha_{22}e^{\frac{\sigma^2}{4\theta}} \right), \quad (3.12)$$

and $\mathcal{R}_0 > 1$ is equivalent to

$$\beta_1 S_1^0(\mu_2 + c_2 + \alpha_{22}) + \beta_2 S_2^0(\mu_1 + c_1 + \alpha_{12}) + \alpha_{12}\alpha_{22} > \beta_1\beta_2 S_1^0 S_2^0 + (\mu_1 + c_1 + \alpha_{12})(\mu_2 + c_2 + \alpha_{22}). \quad (3.13)$$

Additionally, $e^{\frac{\sigma^2}{4\theta}} \rightarrow 1$, $\bar{S}_1 \rightarrow S_1^0$, and $\bar{S}_2 \rightarrow S_2^0$ when $\sigma \rightarrow 0$, which means that (3.12) coincides with (3.13).

4. Numerical simulations

Brucellosis is a zoonotic infectious disease caused by *Brucella* bacteria, mainly infecting domestic animals such as cattle, sheep and pigs, as well as wild ruminants including deer, elk and antelopes. Cross-regional transportation can move live-stock from resource-abundant areas to regions where resources are scarcer but market demand is higher, thereby optimizing resource allocation. However, cross-regional transportation breaks geographical restrictions and expands the range of livestock activities, which may lead to the spread of brucellosis and expose more livestock and people to the risk of infection (Bai et al., 2026). This section takes the transmission of brucellosis between two regions as an example to verify the proposed theoretical results, and also explores the impact of livestock transportation behavior (i.e., migration) and random disturbances on the severity of the epidemic.

4.1. Simulations of the deterministic model (1.2)

In this subsection, we verify the stability of the disease-free equilibrium E^0 and the endemic equilibrium E^* of model (1.2).

The parameters values of model (1.2) are selected as: $m = 3$, $n = 2$, $A_1 = 1200$, $A_2 = 1000$, $\beta_1 = 3.8 \times 10^{-5}$, $\beta_2 = 3.8 \times 10^{-5}$, $\mu_1 = 0.22$, $\mu_2 = 0.22$, $c_1 = 0.15$, $c_2 = 0.15$, $\alpha_{11} = 0.2$, $\alpha_{12} = 0.3$, $\alpha_{21} = 0.2$, $\alpha_{22} = 0.3$, $a_1 = 0.5$, $a_2 = 0.7$. The initial value is $\mathcal{Z}_1(0) = (3500, 60, 10, 10, 10, 2600, 31, 5, 5)$. Under these parameters values, we can easily obtain that $\mathcal{R}_0 \approx 0.5138 < 1$. It then follows from Theorems 2.1 and 2.2 that the disease-free equilibrium $E^0 \approx (5161.3, 0, 0, 0, 0, 4838.7, 0, 0, 0)$ is globally asymptotically stable. Fig. 4.1 shows the curves of S_1 , S_2 , I_1 , I_2 , X_1 , X_2 , X_3 , Y_1 and Y_2 under given initial value, from which we can clearly see that state variables of model (1.2) gradually converge to E^0 over time, and the disease goes extinct in both patches.

To illustrate the persistence of the disease in model (1.2), we choose $\beta_1 = \beta_2 = 1.14 \times 10^{-4}$, while keeping all other parameters and initial value unchanged. The basic reproduction number can be calculated as $\mathcal{R}_0 \approx 1.5415 > 1$, and thus the disease is uniformly persistent from Theorem 2.3. The simulation results are shown in Fig. 4.2 under the given initial value, from which we obtain that the disease persists in both patches and eventually converges to the endemic equilibrium $E^* \approx (3.3659, 1.0692, 1.0692, 1.0692, 1.0692, 3.1197, 1.0204, 1.0204, 1.0204) \times 10^3$.

4.2. Simulations of the stochastic model (1.3)

In this subsection, we numerically illustrate the dynamics for the stochastic model (1.3). The simulation aims to verify the sufficient extinction criterion $\mathcal{R}_0^c < 1$ established in Theorem 3.2, which is formulated solely in terms of model parameters.

We also fix $m = 3$ and $n = 2$ and simulate the stochastic model (1.3) on the time interval $[0, T]$ with $T = 100$. The OU process $x(t)$ is discretized by the Euler-Maruyama scheme, whereas the remaining state variables are updated by the explicit Euler method. The step size is chosen as $\Delta t = 10^{-3}$. We take $\theta = 0.3$ and $\sigma = 0.5$, while all other parameters and the initial value are identical to those in Fig. 4.1. Under this parameter setting, the extinction threshold is

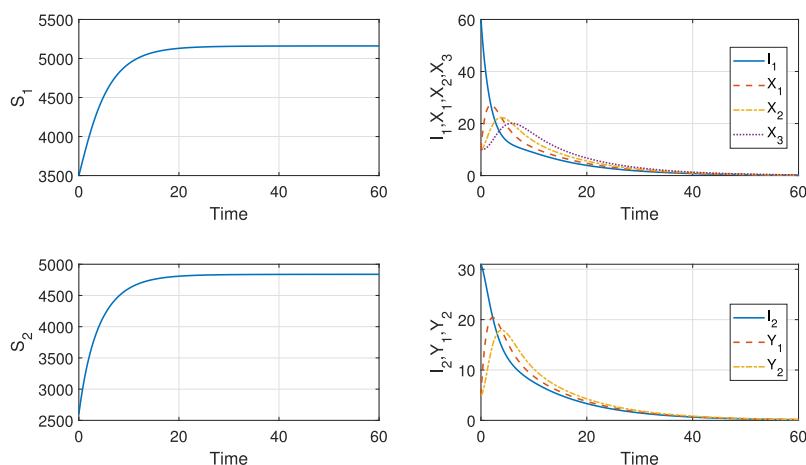


Fig. 4.1. Plots of S_1 , S_2 , I_1 , I_2 , X_1 , X_2 , X_3 , Y_1 and Y_2 versus time when $\mathcal{R}_0 = 0.5138 < 1$.

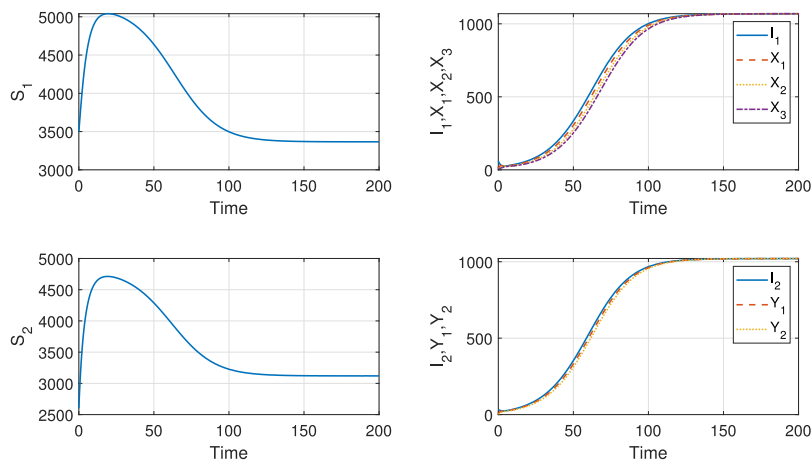


Fig. 4.2. Plots of S_1 , S_2 , I_1 , I_2 , X_1 , X_2 , X_3 , Y_1 and Y_2 versus time when $\mathcal{R}_0 = 1.5415 > 1$.

$$\mathcal{R}_0^e = \max \left\{ \frac{a_1}{\mu_1 + c_1}, \frac{a_2}{\mu_2 + c_2}, \frac{\beta_1 \bar{N}}{a_1}, \frac{\beta_2 \bar{N}}{a_2} \right\} = \max \left\{ \frac{0.35}{0.37}, \frac{0.35}{0.37}, \frac{0.25}{0.35}, \frac{0.25}{0.35} \right\} \approx 0.9459 < 1,$$

so the hypothesis of [Theorem 3.2](#) is satisfied.

[Fig. 4.3](#) confirms the theoretical prediction for $\mathcal{R}_0^e < 1$. The susceptible populations $S_1(t)$ and $S_2(t)$ remain bounded and converge to a disease-free regime, exhibiting only mild stochastic oscillations induced by the OU perturbation (see panels (a) and (c)). In contrast, the infected populations $I_1(t)$ and $I_2(t)$ decay rapidly to zero, and their associated chain variables track the decay with a visible lag due to the Erlang-type distributed delay structure. Specifically, panel (b) shows that $I_1(t) \rightarrow 0$ exponentially and its associated Erlang-type chain variables $X_1(t)$, $X_2(t)$, $X_3(t)$ follow $I_1(t)$ with smoothing and delay; likewise, panel (d) demonstrates exponential extinction of $I_2(t)$ together with the lagged response of $Y_1(t)$, $Y_2(t)$. These sample paths are consistent with the exponential extinction property,

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \ln I_1(t) < 0, \quad \limsup_{t \rightarrow \infty} \frac{1}{t} \ln X_i(t) < 0 \quad (i = 1, 2, 3),$$

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \ln I_2(t) < 0, \quad \limsup_{t \rightarrow \infty} \frac{1}{t} \ln Y_j(t) < 0 \quad (j = 1, 2),$$

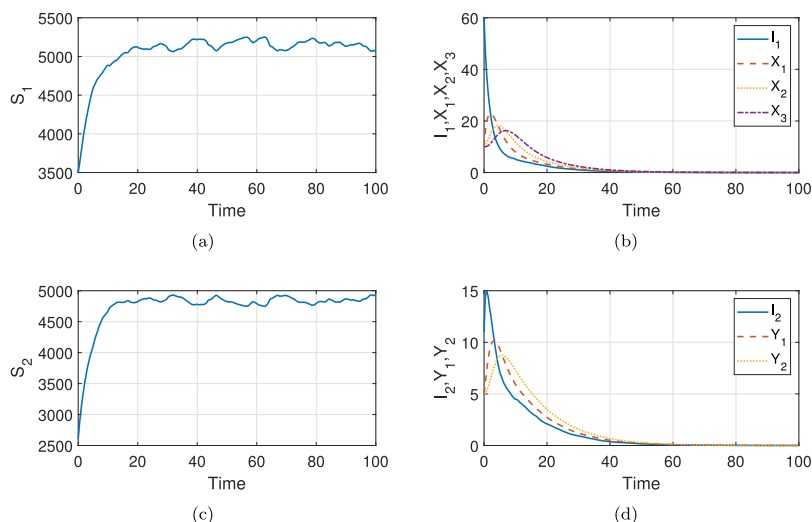


Fig. 4.3. Sample paths of the stochastic model (1.3) under $\mathcal{R}_0^e < 1$.

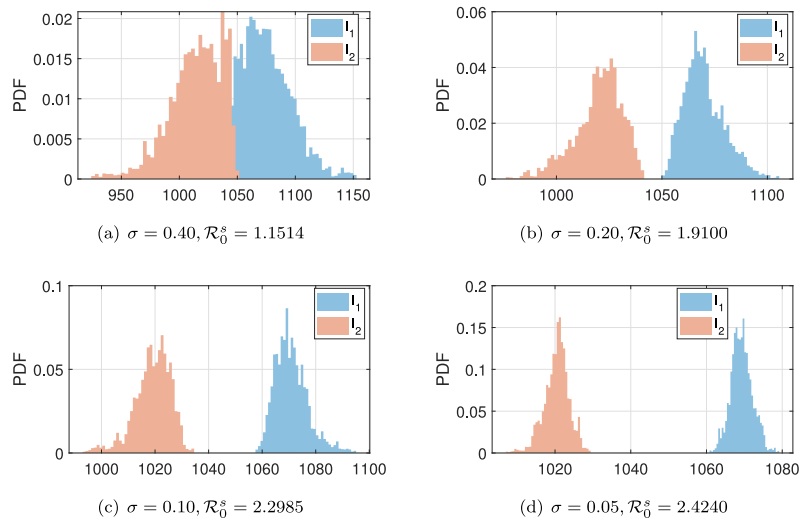


Fig. 4.4. Empirical stationary distributions of the infected populations $I_1(t)$ (blue) and $I_2(t)$ (orange) of model (1.3) when noise intensity $\sigma = 0.40, 0.20, 0.10$, and 0.05 for fixed mean-reversion rate $\theta = 0.1$.

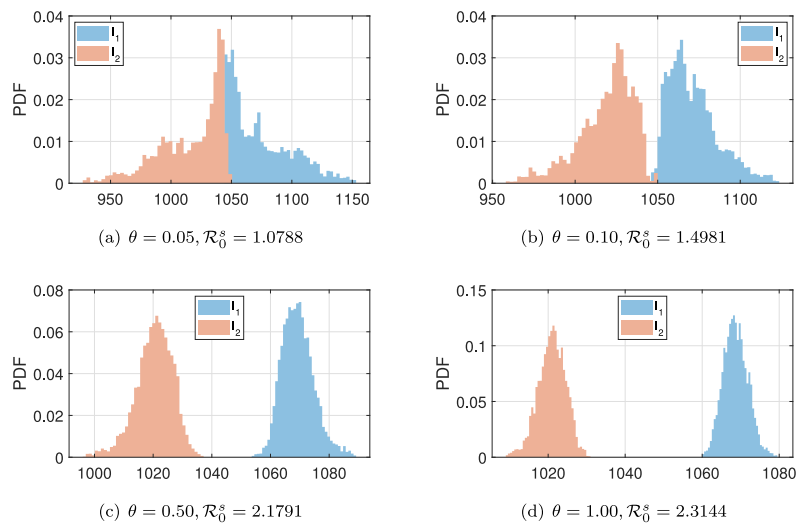


Fig. 4.5. Empirical stationary distributions of the infected populations $I_1(t)$ (blue) and $I_2(t)$ (orange) of model (1.3) when mean-reversion rate $\theta = 0.05, 0.10, 0.50$, and 1.00 for fixed noise intensity $\sigma = 0.3$.

as stated in [Theorem 3.2](#). Overall, the numerical results demonstrate that the infection becomes extinct exponentially fast in the stochastic environment when $\mathcal{R}_0^s < 1$.

We then numerically illustrate [Theorem 3.3](#). The parameter values and the initial value are taken to be the same as in [Fig. 4.2](#). The empirical stationary distributions of $I_1(t)$ and $I_2(t)$, obtained from long-time simulations after discarding a burn-in period, are presented in [Fig. 4.4 and 4.5](#). These figures confirm the conclusion of [Theorem 3.3](#) by demonstrating the existence of non-degenerate stationary regimes when $\mathcal{R}_0^s > 1$, indicating that brucellosis infection will persist under such conditions. Moreover, [Fig. 4.4 and 4.5](#) show that $\mathcal{R}_0^s$ varies with different σ and θ , and the number of infected sheep in both patches also differs accordingly. We will further explore the relationship between σ , θ , and $\mathcal{R}_0^s$ in [subsection 4.3.2](#).

4.3. The impact of transporting behavior and random fluctuations on infection burden

In this subsection, we focus on a scenario involving bidirectional livestock transportation between two farms, analyzing how variations in transporting rate and stochastic disturbance parameters determine the scale of brucellosis outbreaks for deterministic model (1.2) and stochastic model (1.3).

4.3.1. The impact of transporting behavior on infection burden for model (1.2)

The basic reproduction number $\mathcal{R}_0$ is the key threshold determining whether the disease dies out or persists, and the cumulative number of infected sheep directly quantifies the economic loss suffered by farmers. Thus, it suffices to examine how the relative magnitudes of farming scales and transport rates between the two patches—namely the ratios A_1/A_2 , α_{11}/α_{21} and α_{12}/α_{22} —modulate both $\mathcal{R}_0$ and the cumulative infected number. Accordingly, we frame our analysis in terms of these dimensionless ratios rather than the absolute parameter values for the following reasons:

- The absolute scales of two patches fail to accurately capture how one patch influences brucellosis transmission to the other when transporting activities are present. In contrast, relative scale can eliminate the interference of absolute scale values on transmission influence, and reveal differences in transmission risks between patches more clearly, so we introduce the ratio A_1/A_2 . For simplicity, only the case where $A_1/A_2 > 1$ is considered, meaning that Patch 1 is larger in scale than Patch 2.
- When the transporting rates α_{11} , α_{21} , α_{12} , and α_{22} take specific values, their effects on $\mathcal{R}_0$ are shown in Fig. 4.6. From Fig. 4.6(a), it can be observed that when the value of α_{11}/α_{21} remains unchanged, the value of $\mathcal{R}_0$ also hardly varies. Therefore, if α_{11} and α_{21} increase or decrease simultaneously, it will not be possible to effectively study the impact of transporting activities on the prevalence of brucellosis. A similar conclusion can be drawn from Fig. 4.6(b). Hence, it is more reasonable to adopt α_{11}/α_{21} and α_{12}/α_{22} for the study.

We first investigate the simultaneous variations of A_1/A_2 and α_{11}/α_{21} on $\mathcal{R}_0$ and the average number of infections from the initial time 0 to the terminal time $T = 200$ in both patches by keeping α_{12} and α_{22} as constants, which can be shown in Fig. 4.7. From Fig. 4.7(a), it can be observed that increasing α_{11}/α_{21} (i.e., raising the transporting scale of susceptible sheep from Patch 1 to Patch 2) can temporarily reduce $\mathcal{R}_0$. However, as α_{11}/α_{21} continues to increase, the value of $\mathcal{R}_0$ exhibits a rebound. In contrast, Fig. 4.7(b)(c) shows that increasing α_{11}/α_{21} leads to a notable decrease in the number of infections in Patch 1, at the cost of a significant increase in infections in Patch 2 when A_1/A_2 is relatively large (i.e., Patch 1 is significantly larger in scale than Patch 2). This occurs because the transport substantially raises the number of susceptible sheep in Patch 2, thereby multiplying the infection risk. When A_1/A_2 is small, increasing α_{11}/α_{21} has a negligible impact on the infection numbers in both patches. In other words, when there is a substantial difference in scale between the two patches, a moderate increase in the transport of susceptible sheep from the larger patch to the smaller patch can reduce the infection burden in the larger patch without significantly exacerbating the epidemic pressure in the smaller patch.

We now examine the simultaneous variations of A_1/A_2 and α_{12}/α_{22} on $\mathcal{R}_0$ and the average infection numbers in both patches by keeping α_{11} and α_{21} as constants. The results are presented in Fig. 4.8. When A_1/A_2 is relatively large (i.e., Patch 1 is significantly larger in scale than Patch 2), Fig. 4.8(a) shows that increasing α_{12}/α_{22} (i.e., raising the transporting scale of infected sheep from Patch 1 to Patch 2) reduces $\mathcal{R}_0$ to below 1, leading to the eventual elimination of brucellosis in both patches. However, such transport result in a significant increase in the number of infections in Patch 2. Yet, when α_{12}/α_{22} is sufficiently large (i.e., a substantial number of infected sheep are transferred from Patch 1 to Patch 2), the number of infected sheep in Patch 2 gradually declines (since $\mathcal{R}_0 < 1$). Therefore, transporting a large number of infected sheep from Patch 1 to Patch 2 can drive brucellosis toward extinction without significantly increasing the overall infection burden. This strategy is analogous to the practice of collectively isolating and relocating infected individuals during the outbreak of many infectious diseases. When the scale of the two patches is comparable, increasing the transporting rate has a negligible impact on both $\mathcal{R}_0$ and infection numbers. In such cases, alternative measures are required to effectively curb the spread of brucellosis.

4.3.2. The impact of random fluctuations on infection burden for model (1.3)

Fig. 4.4 and 4.5 reveal an interesting phenomenon: as $\mathcal{R}_0^S$ increases, the mean infection level in Patch 1 declines while that in Patch 2 rises. This implies that a larger noise intensity can redistribute the infection burden from one patch to the other. In the following, we further explore this phenomenon and provide a quantitative explanation of how $\mathcal{R}_0^S$ and the noise intensity affect the relative prevalence between the two patches.

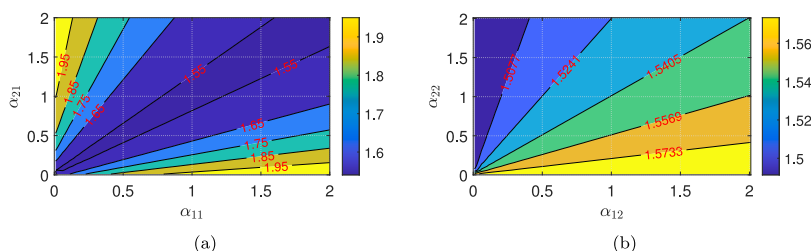


Fig. 4.6. The impact of transporting rates on the basic reproduction number $\mathcal{R}_0$.

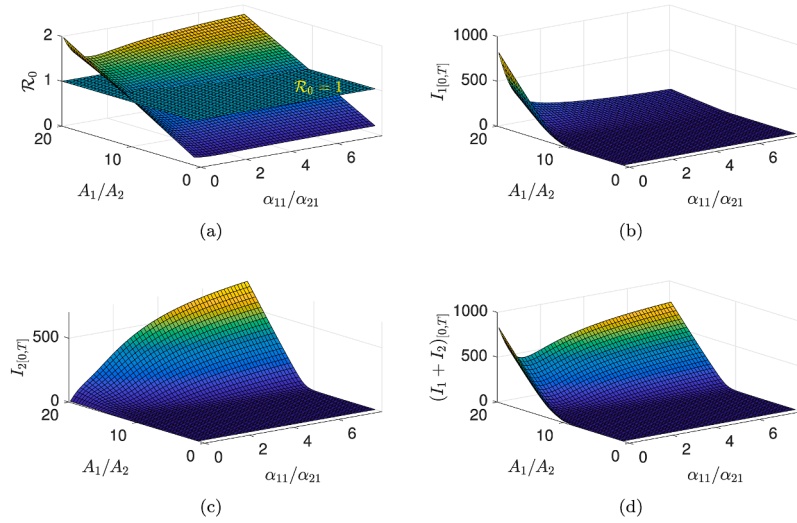


Fig. 4.7. (a) Effects of farming scale and transporting ratio of susceptible sheep on $\mathcal{R}_0$; and (b)(c)(d) infection numbers in two patches. (In (b)(c), $I_\ell[0, T]$ denotes the average number of infected sheep in patch ℓ from initial time 0 to terminal time $T = 200$ for $\ell = 1, 2$; in (d), $(I_1 + I_2)[0, T] = I_1[0, T] + I_2[0, T]$).

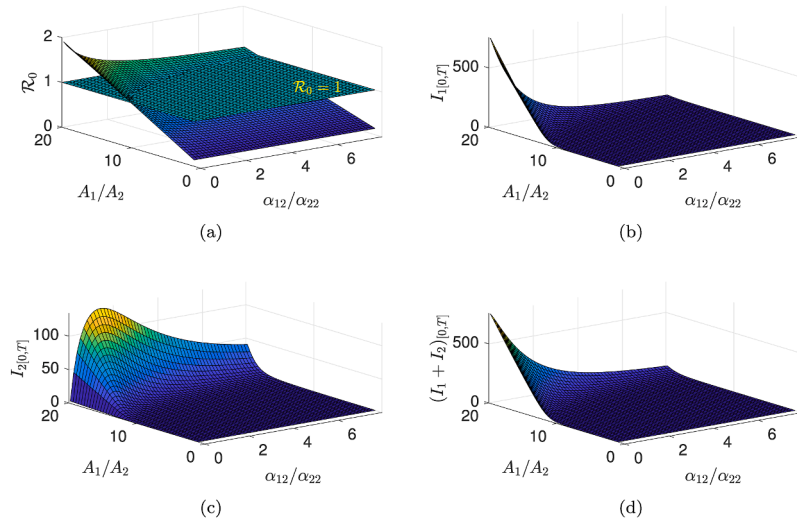


Fig. 4.8. (a) Effects of farming scale and transporting ratio of infected sheep on $\mathcal{R}_0$; and (b)(c)(d) infection numbers in two patches. (In (b)(c), $I_\ell[0, T]$ denotes the average number of infected sheep in patch ℓ from initial time 0 to terminal time $T = 200$ for $\ell = 1, 2$; in (d), $(I_1 + I_2)[0, T] = I_1[0, T] + I_2[0, T]$).

To quantify how the OU-driven transport randomness affects the long-term infection burden of brucellosis, we investigate the dependence of the stochastic threshold $\mathcal{R}_0^S$ and the stationary mean infection levels on the mean-reversion rate θ . In this experiment, we fix the noise intensity at $\sigma = 0.3$ and vary θ over the interval $(0, 1]$, while keeping the remaining parameters and the initial value unchanged. For each θ , the stationary means $\bar{I}_1(\theta)$ and $\bar{I}_2(\theta)$ are estimated from the post-transient segment of a long-time simulation. Fig. 4.9 shows that $\mathcal{R}_0^S(\theta)$ increases monotonically with θ . This behavior is consistent with the fact that a larger θ implies faster mean reversion of the OU process, which reduces the long-lasting effect of environmental perturbations and consequently increases the value of migration-related term in $\mathcal{R}_0^S$. More importantly, as θ increases (and hence $\mathcal{R}_0^S$ becomes larger), the stationary mean $\bar{I}_1(\theta)$ in Patch 1 decreases, whereas the stationary mean $\bar{I}_2(\theta)$ in Patch 2 increases. This observation suggests that a larger noise intensity may shift the infection burden of brucellosis from Patch 1 to Patch 2 through the random transport coupling.

From Fig. 4.10, we observe that the transport-related randomness significantly reshapes the stationary infection burdens across the two patches when $\mathcal{R}_0^S > 1$. In particular, increasing the noise intensity σ or decreasing the mean-reversion rate θ tends to increase the stationary mean infection level $\bar{I}_1(\theta, \sigma)$, whereas $\bar{I}_2(\theta, \sigma)$ exhibits an opposite decreasing trend.

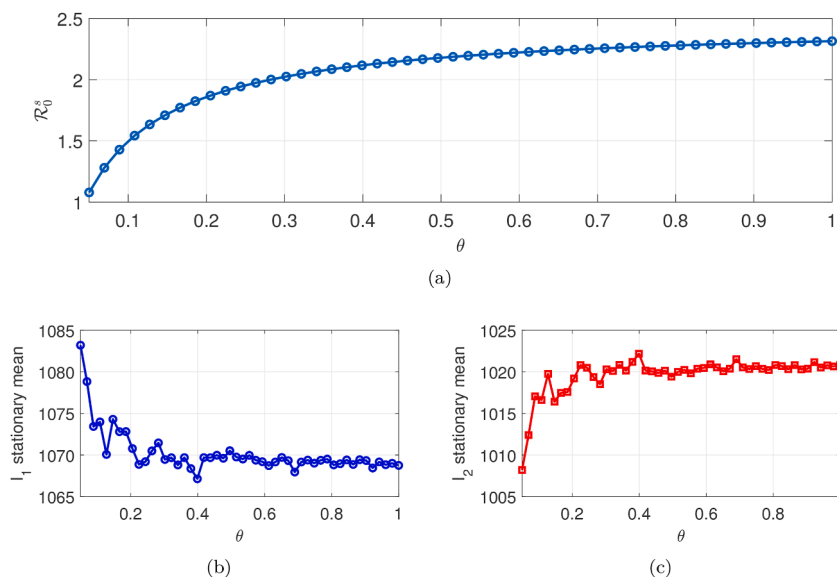


Fig. 4.9. Dependence of the stochastic threshold $\mathcal{R}_0^s$ and the stationary mean infection levels on the OU mean-reversion rate θ , under fixed noise intensity $\sigma = 0.3$.

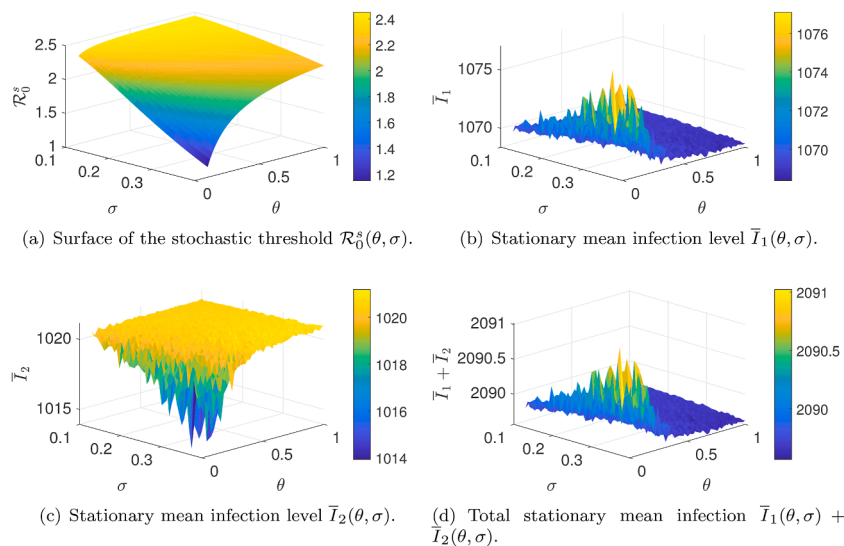


Fig. 4.10. (a) Effects of noise intensity σ and the mean-reversion rate θ on $\mathcal{R}_0^s$; and (b)(c)(d) infection numbers in two patches. (In (b)(c), $\bar{I}_\ell$ denotes the stationary mean of infected sheep in patch ℓ from initial time 0 to terminal time $T = 2000$ for $\ell = 1, 2$).

Moreover, the total stationary mean infection level $\bar{I}_1 + \bar{I}_2$ increases as σ becomes larger or θ becomes smaller, indicating that stronger and more persistent environmental fluctuations amplify the overall endemic burden. Therefore, excessive randomness in transport may aggravate the prevalence of brucellosis even though it redistributes the infection burden asymmetrically between the two patches.

Remark 4.1. The transmission rates β_1 and β_2 of brucellosis in the two patches play a crucial role in the transmission process. Therefore, we numerically investigate the effects of β_1 and β_2 on $\mathcal{R}_0$, $\mathcal{R}_0^s$, and $\mathcal{R}_0^e$, with the results shown in Fig. 4.11. It can be seen from the figure that the magnitudes of β_1 and β_2 have a significant impact on $\mathcal{R}_0$, $\mathcal{R}_0^s$, and $\mathcal{R}_0^e$. More precisely, larger values of β_1 and β_2 lead to larger values of these thresholds, indicating that brucellosis is less likely to go extinct. Furthermore, Fig. 4.11 shows that for the deterministic model (1.2), the curve $\mathcal{R}_0 = 1$ divides the β_1 - β_2 plane into two regions ($\mathcal{D}_1 \cup \mathcal{D}_2$ and $\mathcal{D}_3 \cup \mathcal{D}_4$), corresponding to the extinction and persistence of brucellosis, respectively. However, for the stochastic model, when the parameter σ is nonzero, there exists a gap between the threshold $\mathcal{R}_0^e$ (which determines extinction)

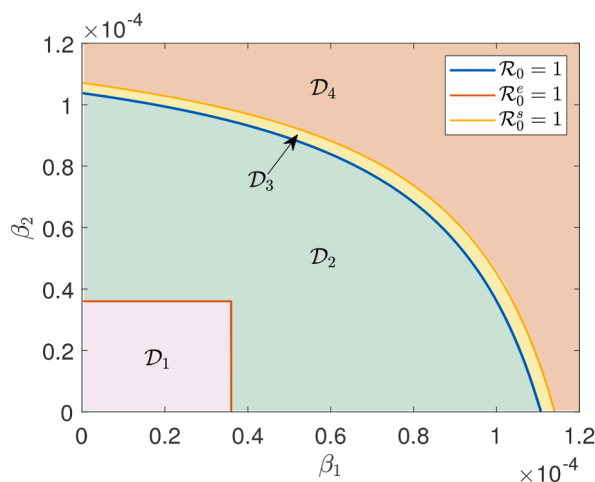


Fig. 4.11. Comparison of $\mathcal{R}_0$, $\mathcal{R}_0^e$, and $\mathcal{R}_0^s$ (the regions where $\mathcal{R}_0 < 1$ are $\mathcal{D}_1$ and $\mathcal{D}_2$; the regions where $\mathcal{R}_0 > 1$ are $\mathcal{D}_3$ and $\mathcal{D}_4$; $\mathcal{R}_0^e < 1$ in region $\mathcal{D}_1$, and $\mathcal{R}_0^s > 1$ in region $\mathcal{D}_4$).

and the threshold $\mathcal{R}_0^s$ (which determines the existence of a stationary distribution). Narrowing this gap will be one of the directions for our future work.

5. Conclusion

In this paper, we investigated the global dynamics of a two-patch epidemic model that integrates spatial migration, distributed delays, and environmental randomness. The proposed framework extends classical multi-patch epidemic models in a mathematically tractable yet biologically meaningful manner. In particular, the infection process is assumed to involve temporally distributed progression, represented by Erlang-distributed delays and converted into a finite-dimensional system by means of the linear-chain technique. Meanwhile, random fluctuations in migration-related interactions are introduced via an Ornstein–Uhlenbeck (OU) process, capturing mean-reverting environmental perturbations that naturally arise in spatially coupled populations. This combination enables the model to simultaneously describe stage-structured infection mechanisms, spatial coupling, and persistent environmental variability.

For the deterministic model (1.2), we established threshold-type results that completely characterize both disease extinction and persistence. By analyzing the invariant region and employing comparison principles, we proved that the disease-free equilibrium is globally asymptotically stable whenever the basic reproduction number satisfies $\mathcal{R}_0 < 1$, whereas the infection remains endemic when $\mathcal{R}_0 > 1$. These results rigorously confirm that distributed delays preserve the classical threshold principle, while enriching transient dynamics by introducing delay-induced smoothing and time-lag effects. For the stochastic model (1.3), our analysis demonstrated that migration-related environmental noise may fundamentally reshape the long-term infection dynamics. First, we derived explicit sufficient conditions for almost sure exponential extinction of infection. This was achieved by constructing suitable Lyapunov functions for the infected subsystem and exploiting its Metzler structure, which enables the stochastic perturbation to be handled systematically. The extinction result provides a parameter-only criterion that ensures $\limsup_{t \rightarrow \infty} \frac{1}{t} \ln I_1(t) < 0$ and $\limsup_{t \rightarrow \infty} \frac{1}{t} \ln I_2(t) < 0$, almost surely, together with the exponential decay of the associated Erlang-type chain variables. Second, motivated by the notion of stochastic persistence, we proved that $\mathcal{R}_0^s > 1$ guarantees the existence of a stationary distribution for the stochastic model. This invariant measure can be interpreted as the stochastic analogue of the endemic equilibrium in the deterministic setting and provides a rigorous description of persistent random oscillations around endemic levels. Extensive simulations corroborated and refined the following conclusions: the infection dies out exponentially under the derived sub-threshold conditions, whereas above the threshold the densities converge to a stationary distribution. Moreover, numerical surface plots in the (θ, σ) -plane revealed that both the invasion potential (through $\mathcal{R}_0^s$) and the stationary infection burden are sensitive to the magnitude and temporal persistence of the OU perturbation. Notably, varying the noise intensity σ and the mean-reversion rate θ can produce nontrivial redistribution of the long-term infection burden between patches, highlighting how migration-linked randomness fundamentally reshapes spatial epidemic patterns. The mathematical framework developed in this work is highly versatile and can be extended to a variety of epidemic scenarios involving spatial heterogeneity, delayed transmission mechanisms, and random environmental effects. Based on this, several promising research directions are worthy of further investigation (Li et al., 2025a, 2025b).

On the theoretical side, it would be of interest to extend the present framework to stochastic partial differential equation (SPDE) models incorporating spatial diffusion (Pájaro et al., 2025; Su et al., 2025). In such infinite-dimensional settings,

establishing sharp threshold conditions for stochastic persistence and extinction becomes considerably more challenging due to the interplay between spatial heterogeneity and environmental noise. Developing a rigorous threshold theory for stochastic reaction–diffusion epidemic systems therefore remains an important open problem. Meanwhile, further study of the existence, uniqueness, and ergodicity of invariant measures for the associated SPDE system is of significant interest. Deriving quantitative convergence rates toward the invariant measure would provide deeper insight into the long-term dynamics and clarify how spatial diffusion and stochastic perturbations jointly affect epidemic evolution. On the other hand, the Erlang distribution offers a flexible and analytically tractable approach to model distributed delays by exploiting the linear chain trick, but it is important to acknowledge that the choice of delay kernel can influence model dynamics. Future work could explore the impact of alternative kernels, such as the Dirac delta for fixed discrete delays or more general gamma distributions, which may be more appropriate for capturing specific biological waiting-time processes (e.g., fixed incubation periods vs. variable infectious periods).

On the modeling and computational side, the current two-patch framework can be naturally extended to spatially continuous or network-based structures, where population movement is governed by diffusion operators or complex contact networks. Incorporating regime-switching environments, multi-source stochastic perturbations, or correlated noise processes would allow the model to capture more realistic environmental and socio-economic variability (Zhu & Yin, 2009). Moreover, coupling OU-driven stochastic migration with reaction–diffusion mechanisms may provide a unified framework to study the interplay between spatial spread and environmental randomness.

From an applied perspective, another meaningful direction is the development of data-driven parameter estimation and model calibration techniques (He et al., 2010; Ionides et al., 2006). Integrating mobility data, livestock transportation records, or epidemiological surveillance datasets could improve the estimation of both epidemiological parameters and stochastic migration intensity. Such efforts would enhance the predictive capacity and practical applicability of stochastic delayed multi-patch models in livestock disease control and other spatially coupled infectious disease systems influenced by environmental and socio-economic variability.

These extensions are expected to broaden the theoretical scope of stochastic epidemic modeling while simultaneously strengthening its practical relevance in real-world disease management.

CRedit authorship contribution statement

Ting Kang: Writing – review & editing, Writing – original draft, Formal analysis. **Boqiang Cao:** Writing – review & editing, Software, Methodology, Conceptualization. **Zhenfeng Shi:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology. **Qingyun Wang:** Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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